

Sharing the Gains of Aggregation

Cooperative Redistribution of Imbalance Cost Savings

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Abstract

As renewable penetration increases, Balance Responsible Parties (BRPs) face growing uncertainty in matching supply and demand. While portfolio aggregation measurably reduces both imbalance costs and risk, it also obscures each consumer or producers true contribution, particularly as TSO's in the Nordic countries move towards partially asymmetric pricing in the *full-cost balancing* imbalance settlement scheme.

This thesis develops a cooperative-game-theoretic framework to allocate ex-post imbalance costs and risk among prosumers in a 22-client portfolio. Only the day-ahead and imbalance markets are considered, and an asymmetric two-price imbalance settlement is used to approximate full-cost balancing. Cost and risk allocation are modeled as transferable-utility games. Multiple core-selection mechanisms are implemented: The Shapley Value, VCG, the Gately Point, Nucleolus, and two bespoke mechanisms: Uniform Price and Reduced Cost. These are compared against a flat rate allocation where the BRP makes no effort to differentiate between clients.

Each mechanism is assessed for its computational tractability, fairness, budget balance, and incentive compatibility. The Shapley Value and Nucleolus are found impractical for these problems, as they are too computationally taxing to scale for large portfolios. Likewise, VCG is impractical because of its lack of budget balance. A modified VCG mechanism that ensures budget balance for this problem is proposed, however, it results in unstable allocations for some games.

Among the classical core selection mechanisms, the Gately Point proves to be the only practical mechanism, based on computational complexity, stability and budget balanced. The Reduced Cost and Uniform Price mechanisms prove practical and fair, and can be considered as simpler alternatives to the Gately Point.

While several tested mechanisms yield stable allocations, each inherently favors certain types of clients. In some cases, a client's allocated cost may more than double when transitioning from a mechanism that benefits them to one that does not. As forecast accuracy improves, these disparities become more pronounced, making it increasingly important to understand which clients are advantaged or disadvantaged by a given mechanism.

The application of cooperative game theory to BRP pricing is found to be an important consideration, as there are significant differences in cost and risk between clients in the portfolio. Providing similar rates for all clients distributes costs unfairly and does not reward clients for having low imbalances. Therefore, cooperative game theory can aid prosumers by assigning costs that reflect their balancing needs, and aid BRPs and the electrical grid by incentivizing prosumers to work towards predictable consumption.

Preface

This thesis was prepared at the department of Wind Energy at the Technical University of Denmark in fulfillment of the requirements for acquiring a M.Sc. degree.

The research described in this thesis is the result of a cooperation between Reel Energy ApS and DTU Wind. The initial idea behind the cooperation was to develop a model for controlling a battery that was shared between a portfolio of clients as part of Reel's *Helios* project. After considering this problem, the issue of how to properly allocate the gains from the battery among shareholders, seemed more interesting and outside of Reel's expertise, and raised questions on how gains of cooperation could be shared in other parts of Reel's business.

Lyngby campus, Lyngby, August 22, 2025

A handwritten signature in blue ink, appearing to read 'Asmus', followed by a long, sweeping horizontal stroke.

Asmus Winther Eriksen

Acknowledgements

When I first began thinking about my thesis, all I knew was that I wanted it to connect with the real world. After years immersed in theory, I was burned out on purely academic pursuits.

Collaborating with Reel on this project has shown me that research can be both interesting and impactful. It has reignited my curiosity and reminded me why I love learning.

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CHAPTER 1

Introduction

1.1 Use of artificial intelligence

Artificial intelligence has been utilized in the creation of the code for the project and the readme file.

When creating a complicated function, the function was first written in a way that was simple and fast to create, and then AI was used to improve it while keeping the functionality, if needed. E.g. the characteristic function calculation and nucleolus implementation were initially written using many dictionaries and inefficient loops, but made faster using AI.

Simple functions like plotting and data manipulation have in some cases been created entirely by AI.

AI has been used to format tables and figures with author created content. E.g. creating subfigures, switching rows in tables, changing spacing.

AI has also been used to create first drafts of abstract, conclusion, and nomenclature.

AI has not been used in any other way related to creating the content of the thesis.

1.2 The problem

The problem studied in this thesis is balancing the amount of power purchased and consumed, and how the portfolio effect influences that. When any consumer uses electricity, it must be purchased in a series of markets. It is a balancing responsible party's (BRP) role to ensure that consumption and purchased energy on these markets match up.

It is important that produced and consumed power always match in a power system and when there is a discrepancy between these, the imbalance price that reflects the cost to remediate this must be paid. Imbalances can be either in the "dominating" or "dominant" direction or in the non-dominant direction. An imbalance in the dominant direction has the same direction as the overall system imbalance, so it contributes to increasing the imbalance in the system. An imbalance in the non-dominant direction is opposite, and so contributes to reducing the imbalance.

In a one-price system the cost is symmetrical. Agents with imbalances in the dominant direction, called "polluter", will be penalized the same per unit of imbalance as agents with opposite imbalances are rewarded. In the Nordic countries, the Transmission System Operators (TSO's) are moving towards a *full cost balancing* system [15]. The reason for this change is that procurement of reserve balancing capacity that can be activated when imbalances aren't resolved in the preceding markets, is expected to rise by 50% in DK1 and 33% in DK2 by 2030 [15]. The current one-price imbalance system only covers

the costs of activation of balancing units, but the reserve capacity cost is covered by the TSO, and in the end by regular consumers.

The purpose of full cost balancing is to make polluters cover the entire cost of their imbalance. The penalty will only apply to imbalances in the dominant imbalance direction [15], and therefore this is a move towards a system with two prices. The dominant imbalance direction will be the direction of the overall system imbalance. In anticipation of this, the imbalance market in this thesis will be modelled as a two-price scheme, with polluters paying the imbalance cost in the dominant direction and the spot price in the non-dominant direction.

In this thesis, a BRP that is remunerated based on spot price plus a flat fee per unit of energy used is studied. This form of payment means that the BRP isn't directly exposed to the spot price, instead they are exposed to the *imbalance spread* λ . Which is the difference between imbalance price $\lambda^{Imbalance}$ and spot price λ^{DA}

$$\lambda = \lambda^{Imbalance} - \lambda^{DA} \quad (1.1)$$

The imbalance spread is the difference between the imbalance price and the day ahead price. Suppose the imbalance price is $\lambda^{imbalance} = 100 \frac{EUR}{MWh}$ and the day ahead price is $\lambda^{DA} = 50 \frac{EUR}{MWh}$. If the BRP buys $1MWh$ too little, they will have to pay 100€ for the imbalance, but the consumer pays the spot price of 50€, so the total cost to the BRP is 50€.

In a one-price system, this is symmetrical. So a BRP that bought $1MWh$ too much would get paid 50€, but for this thesis a two-price system is assumed where polluters pay and others do not get paid, in line with the preliminary full cost balancing model [15]. This means that the spread would be $50 \frac{EUR}{MWh}$ for buying missing electricity and $0 \frac{EUR}{MWh}$ for selling excess electricity.

When aggregating multiple clients in a portfolio, the portfolio effect will reduce the variance of the imbalance. This comes as a result of forecasting errors counteracting each other. A simple example can be seen in figure 1.1. Three clients are in the portfolio, each have an imbalance caused by buying an inaccurate amount of electricity for them. By aggregating them, the overall imbalance is reduced. Assuming the imbalance spread is $50 \frac{EUR}{MWh}$ with the positive direction being dominant, the unaggregated cost would be:

$$1MWh \cdot 50 \frac{EUR}{MWh} + 2MWh \cdot 50 \frac{EUR}{MWh} - 2MWh \cdot 0 \frac{EUR}{MWh} = 150EUR$$

while the aggregated cost in a two-price system would be

$$1MWh \cdot 50 \frac{EUR}{MWh} = 50EUR$$

In figure 1.2, the costs of a portfolio can be seen in a two-price system, assuming that client imbalances are normally distributed. The two-price system will set all positive income to 0. Therefore; the expected income will be lower than the mean.

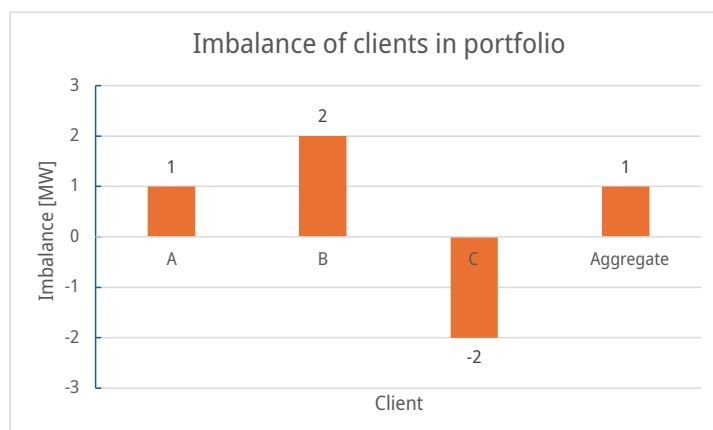


Figure 1.1: Effects of aggregating clients for a single imbalance realization

When summing normal distributions, the mean μ of the combined distribution will be the sum of the other distributions $\mu_{sum} = \mu_1 + \mu_2$, while the standard deviation σ of the combined distribution will be $\sigma_{sum} = \sqrt{\sigma_1^2 + \sigma_2^2}$, resulting in the combined portfolio having a lower standard deviation than if the two were separate clients.

In a one-price scheme, since the mean would be the same, the average cost of having the clients in separate portfolios and having them combined would be the same, but in the two price scheme, the lower standard deviation causes the expected income to be higher when aggregating clients.

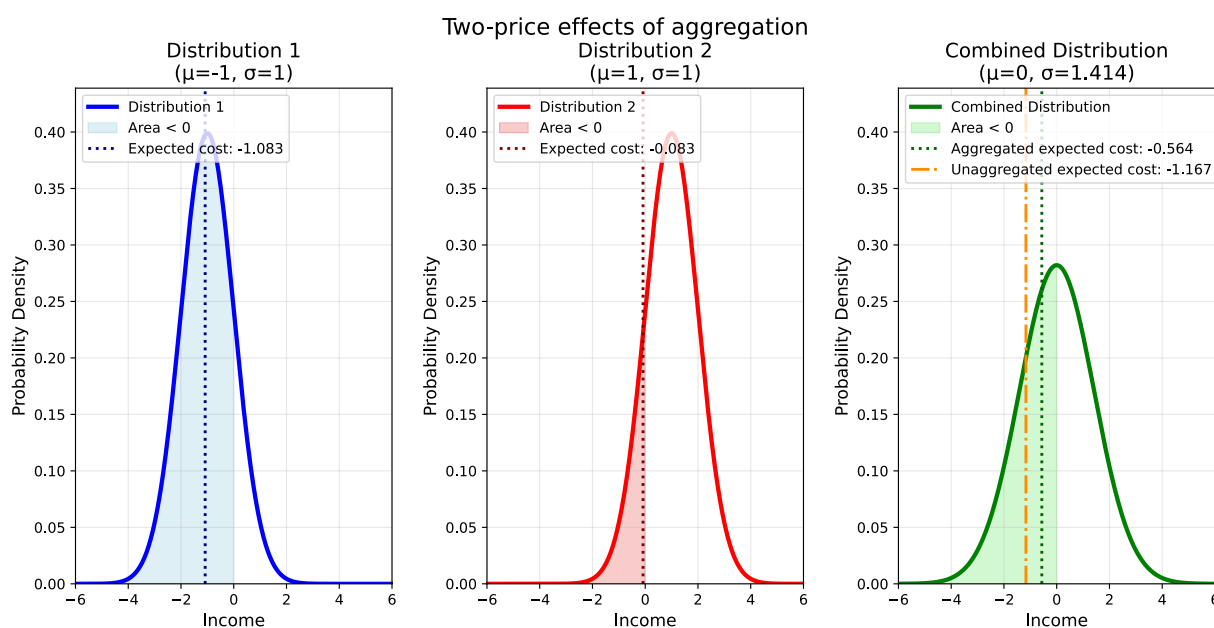


Figure 1.2: Effects of aggregating clients over time in a two-price system. Distribution indicates one-price income, area indicates two-price income

In a one-price system, aggregation would not reduce costs as the expected cost would be the same as the mean of the distribution, and the means are simply added together, however; aggregation is still beneficial in a one-price system.

In figure 1.3, the effects of aggregation on risk of high costs can be seen. The conditional value at risk (CVaR) with $\alpha = 0.1$ is used as the risk metric in this comparison, indicating the average income in the 10% worst scenarios. CVaR is defined and described more in depth in section 3.2.2.

From figure 1.3, it is apparent that CVaR is increased as the clients are combined. Adding client 2 to the portfolio reduces CVaR, rather than increasing it. The results from aggregating clients on CVaR will be the same for two-price and one-price schemes, as long as α is chosen so no client has a positive income in any of their worst case scenarios.

This thesis will study how aggregation of clients impact the cost and risk of a portfolio

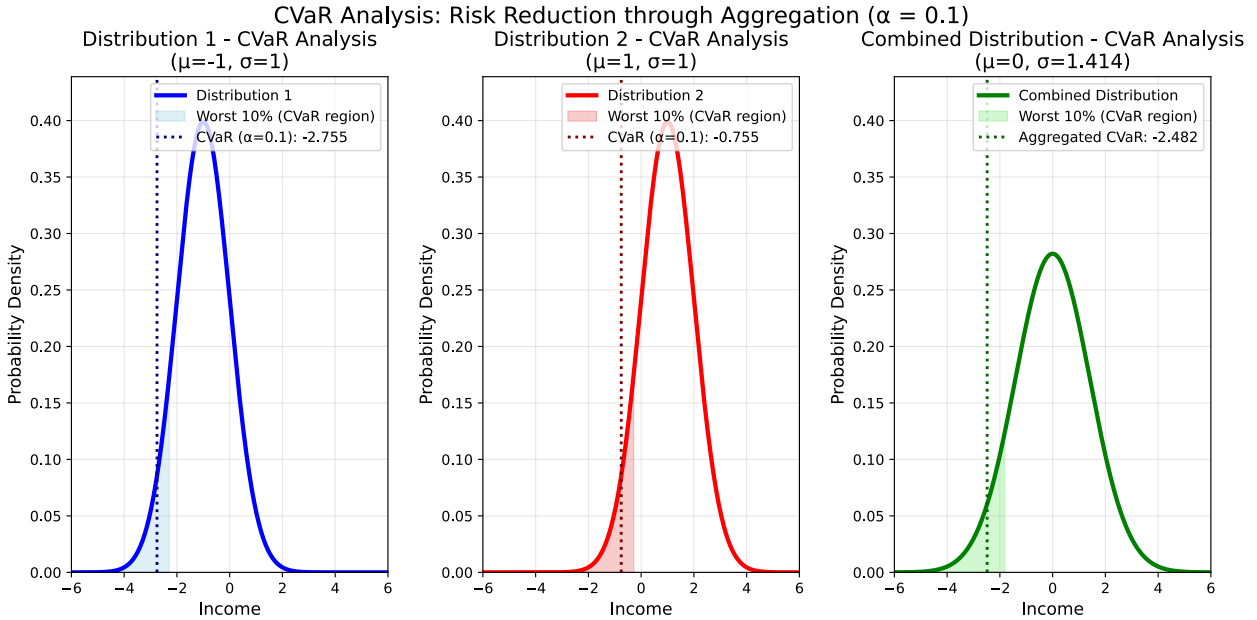


Figure 1.3: Effects of aggregating clients over time on CVaR

in a two-price system, by modeling it as a cooperative game where the goal is to minimize costs.

By properly quantifying the cost and risk, it will be possible to make more fair agreements between BRP's who buy the electricity and consumers who consume it, that properly reflect the cost of the consumers imbalance. The exact usage of this data can vary depending on the market. The most similar to the current market conditions would be to receive historical data from the clients, and set fees ex-ante that reflect the expected imbalance costs incurred by the client.

Another possibility is a model as seen from insurance companies, where a relatively high fee is agreed upon ex-ante, but the clients are compensated by a rebate when actual

imbalances incurred are calculated ex-post. Similarly, a payment on account system as sometimes seen with power bills could be implemented where the deviations from the fee, either positive or negative, are resolved ex-post.

The actual implementation of this system of fair payment can be difficult, as the purpose of a BRP is to assume the risk of imbalances. In a system where the client is more directly exposed to the costs of their imbalances, the client assumes risk that they have limited control over, so this is likely to be a product with lower cost for risk taking clients.

In summary, consumers cause imbalances in produced and consumed electricity in the electric grid. It is the role of BRP's to take responsibility for this imbalance, but correctly attributing the cost and risk each consumer adds cannot be done by looking at only the individual consumer, as the whole portfolio will influence the cost. This thesis will study the best ways to attribute this cost and risk to individual consumers, so it can be applied to make competitive agreements that reflect true costs. This leads to the central research question:

How can the true imbalance costs of prosumers in a portfolio be allocated using cooperative game theory?

1.3 The Company Case

This thesis examines the case of Reel Energy ApS, a Danish broker specializing in Power Purchase Agreements (PPAs) and Balance Responsible Party services. The focus is on their 25-signatory PPA centered around in Mesballe, Denmark [4]. As data is only available for 22 signatories, see section 6.2, only 22 will be studied in this thesis. This agreement is notable for its high number of signatories, each of whom has acquired a share in a 14 MW photovoltaic (PV) plant. The signatories represent a diverse mix of customer types and demand profiles [4], resulting in a heterogeneous portfolio. This diversity offers a robust foundation for analyzing how risk and cost may vary across different customer segments.

1.4 Literature Review

Since the introduction of the Shapley Value in 1951[34], research in the basic theory behind cooperative game theory has continued, with concepts such as the Vickrey-Clarke-Groves mechanism being developed 1961-1973 [41][10][23], the nucleolus in 1969[33], and the Gately Point being introduced in 1974[19]. Beyond introducing the Gately Point, Gately's paper is notable as studying investment planning in electric power generation, making it one of the earliest works combining cooperative game theory and energy economics.

Since then, cooperative game theory has become widely noted, as seen by Shapley's Nobel Prize in Economic Sciences from 2012. It has been especially adopted in energy

economics, and has been applied to many cases within the field. Some examples are: Sharing the gains from energy communities [25], sharing energy storage capacity [40], exploring alternatives to marginal pricing in electricity markets [16].

Much of the existing research has focused on the supply side and microgrids, but as markets become more liberalized, it is pertinent to look at the demand side of the market as well. Srinivasan et. al. [37] study how price-maker buyers can form coalitions and bid strategically in order to increase profits. Ayón et. al. [1] studied how price-taking aggregators should bid in day-ahead and spot markets, focusing on the optimization and generation of the bids, but not employing game-theory to redistribute the gains.

As the share of renewables rise, the uncertainty in energy markets rise as well [8]. An aspect in countering this is aggregation of diverse assets, contributing to the portfolio effect and reducing uncertainty. Investing in different and geographically spread generation technologies to reduce uncertainty from the supply side is widely studied [18] [39]. Baeyans et. al. specifically employs cooperational game theory to reduce uncertainty and increase profits of wind turbines through aggregation [2]. On the demand side, diversification in terms of contract types has been studied, showing that it is preferable to have a mix of bi-lateral contracts and spot market exposure [21] and that hedging both in terms of power price and generation sources is preferable [5].

Demand side uncertainty can be mitigated partly by bidding curves, anticipating the demand of price-elastic clients [36], but as ownership of consumer power supply increases, demand side uncertainty increases further [31]. Russo has found that as consumer renewables become more common, the risk to retailers rise, driven more by the load risk from intermittent self sufficiency, than price risk [31].

One approach to countering this is to calculate time-of-use tariffs for all clients, that clients pay based on the uncertainty in that usage time [29]. This approach is fitting if individual client uncertainty is not a significant factor, e.g. for large portfolios, but doesn't differentiate between clients.

For a retailer with a low amount of clients that each carry varying amounts of risk, it might be useful to be able to attribute their risk and cost as part of the portfolio to them. Cooperative game theory can be applied to this problem, quantifying each clients contribution based on fair principles, enabling the retailer to make intelligent decisions regarding their portfolio and pricing.

CHAPTER 2

Theory

2.1 Game Theory

Game theory is the study of interactions between rational and intelligent agents. In this context, rational means that agents do what is best for them, they will choose the course of action that maximizes their own game. Intelligent means that the agents have the means to understand the problem and find the best strategies[27].

A natural example of a game is chess. The agents each are rational because they want to win the game, and they are assumed intelligent because they know all the moves and counter moves. In the real world, the assumption of agents being able to compute every single move is improbable, but intelligence must be assumed when designing the game of chess, as fairness of the game is important.

2.1.1 Theory of cooperative games

When studying game theory and decision making, games can be divided into cooperative and non-cooperative games. The main distinction being that in cooperative games, obligatory contracts between agents are possible, while they are not in non-cooperative games [6].

A classic example of a game is the prisoners dilemma. Two agents can choose whether to turn the other in, or to stay silent. There are three possible outcomes: If no agent turns the other in, both will get a mild sentence. If one agent turns the other in, the snitching agent gets freed and the other agent gets a harsh sentence. If both agents turn each other in, they will both get a medium sentence.

In a non-cooperative game, the agents do not trust each other, and assume that the other agents will make the choice that maximizes their personal gain. This leads to the dominant strategy being to turn the other agent in, leading to both agents getting a harsh sentence.

In a cooperative game, the agents can make an agreement beforehand that they will work together, leading to the collectively best outcome of not turning each other in.

The prisoners dilemma is a simple game, but introducing complications can make it more interesting to study. Maybe one agent has committed a harsher crime than the other, leading to them benefiting more from not being turned in. In this case, the agents may make a deal that one agent pays the other, in order to maintain fairness.

The problem studied here, will be studied as a set of cooperative games. This allows agents to collude in order to get the best possible outcome for the group as a whole, as

long as the group acts truthfully [3].

A game can be further described as having transferable utilities if money or something else can be used to transfer the utility between participant in a manner that incentivizes effective cooperation. The prisoners dilemma can be a transferable utilities game if the prisoners are able to pay each other to compensate.

Games can be maximization or minimization games. Many cooperative games are described as maximization games, and much of the literature defines terms based on those. The games in this thesis are defined as cost reduction games, with a goal of minimizing expenditure. Cost reduction games have an equivalent savings game, where the purpose is to maximize the savings compared to the baseline. An example is the prisoners dilemma. It can be described as a minimization game, where both agents want as low a sentence as possible, or it can be described as a maximization game, where both agents want to maximize their reduction in their sentence. The same theory applies to minimization games and maximization games with slight changes in terminology or signs. In this thesis, the games are kept as cost reduction games, so minimization terms are used instead of maximization terms throughout the report.

In cooperative games, **agents** $i \in N$ and **coalitions** S are central terms. With agents being the individual clients in the set of all agents N and coalition being a subset of N , $S \subseteq N$, that could decide to work together. If a coalition is made up of all the agents it is called the grand coalition N , if it is made up of no agents, it is the empty set $\emptyset$. Together, the agents participate in a game v . The game is a set of **characteristic functions** $v(S)$ which defines the maximum utility a coalition can produce. The general purpose of a minimization game is to maximize the utility by minimizing the total cost for all players, by assigning them to coalitions.

In the context of the prisoners dilemma, the characteristic function would be the sentence length. Suppose a short sentence is 1 year, a medium is 3 years and a long is 5 years. If a prisoner acts alone, they can expect the other agent to turn them in, and therefore their best choice is to turn the other agent in for a medium sentence: $v(\{1\}) = v(\{2\}) = 3$ for a collective sentence of 6 years. If they work together, they can collectively achieve a shorter sentence: $v(N) = 2$. As this is a total of 4 years shorter than each agent working alone, the agents will have an incentive to work together.

An important property of some maximization cooperative games is **super-additivity**, meaning that the value of two coalitions working together will be at least as much as their value working apart $v(S_1) + v(S_2) \leq v(S_1 + S_2)$. The minimization equivalent to this is **sub-additivity** where $v(S_1) + v(S_2) \geq v(S_1 + S_2)$. This guarantees that it will be beneficial in regards to total utility to form the grand coalition. The prisoners dilemma is a sub-additive game, as the total sentence decreases when the agents join a coalition together. An **inessential** game is a game where $v(S_1) + v(S_2) = v(S_1 + S_2)$, so there is no additional value being created by cooperation.

A stronger condition than super-additivity is convexity, which requires the marginal contribution of each individual agent to increase as coalition size increases [6]. The minimization equivalent is concavity and holds if the following is true:

$$v(S_1 \cup S_2) + v(S_1 \cap S_2) \leq v(S_1) + v(S_2), \quad \forall S_1, S_2 \subseteq N \quad (2.1)$$

The sum of the cost of two coalitions should be greater or equal to the sum of the cost of the union + the intersection of them. This means that the marginal contribution of a player to any coalition should be lower than their marginal contribution to any subcoalition of that coalition. So a player is always worth the most as part of a larger coalition. Just like sub-additivity, concavity is useful for a minimization game because it facilitates working together.

Note that that concavity or convexity of the game is not the same as concavity or convexity of the underlying optimization problem. For the problem in this thesis, the characteristic functions are found partly by an optimization problem, but the convexity or concavity of the optimization problem does not influence the concavity of the game. The characteristic function defines the total utility available to the group, but does not say anything about how to divide this utility. Sub additivity for minimization games ensures that the grand coalition will be desirable for the group as a whole, but not necessarily for each individual agent. How to divide the utility is the central question in this thesis, and many tradeoffs must be taken into account to decide this. This division can be described in an **allocation vector** $x^S = \{x_i^S \mid i \in N\}$, where x_i^S is the payments attributed to agent i from the total cost of coalition S .

For an alternative prisoners dilemma where the agents are given a fine instead of a jail sentence, the agents might decide among themselves to pay each other a portion of the fine. If one prisoner has committed a worse crime than the other, they might pay a portion of the other agents costs in order to not get turned in.

In order for it to be sensible for an agent to participate in a coalition, **individual rationality** must hold. Meaning that the payments of each agent in a coalition, cannot be higher than when working alone: $x_i^S \leq v(\{i\})$. If costs were lower when working alone, the agent is likely to leave the coalition. Suppose a prisoners dilemma with fines has the following characteristic functions: $v(\{1\}) = 500$, $v(\{2\}) = 1000$, $v(N) = 1200$. Agent 1 will not agree to paying more than 500 of the collective fine, as in that case the agent would be better off working alone and turning the other agent in.

Collective rationality must also hold, the total payments equals the total cost of the coalition $\sum_{i \in S} x_i^S = v(S)$. If a payment vector satisfies these conditions, it is an **imputation**. The prisoners cannot decide to pay more or less than their collective fine. The set of imputations can be described by a series of inequalities. For the prisoners dilemma with fines described above, the imputations can be described with: $x_1^N \leq 500$ and $x_2^N \leq 1000$ from individual rationality and $x_1^N + x_2^N = 1200$ from collective rationality. Any division of costs not in this set will not be feasible.

In order to measure the dissatisfaction of a coalition S_1 with the payoff as part of a different coalition S_2 where $S_1 \subseteq S_2$, the **excess** for a minimization game is defined:

$$\epsilon(S_1, x^{S_2}) = \sum_{i \in S_1} x_i^{S_2} - v(S_1) \quad (2.2)$$

[25]. Meaning the difference between the utility generated by the coalition working alone, and the payoffs given to the coalition as part of another coalition. If the excess is positive, the coalition pays more as part of S_2 than they would in S_1 , and would therefore prefer to leave S_2 . If an imputation results in all coalitions having a non-positive excess with

respect to the grand coalition, it is part of the **core** that is a subset of the imputations. An allocation in the core will be an allocation where no subset of the grand coalition is better off splitting off and working alone. While an imputation is fair for all agents with respect to working alone, only imputations in the core ensure long term stability of the grand coalition.

Expanding the prisoners dilemma to a group of three agents, the characteristic functions could be: $v(\{1\}) = 500, v(\{2\}) = 1000, v(\{3\}) = 2000, v(\{1, 2\}) = 1200, v(\{1, 3\}) = 2100, v(\{2, 3\}) = 2500, v(N) = 3000$. Now the agents have the possibility of forming coalitions that are not the grand coalition or singleton coalitions. A possible imputation is $x_1^N = 500, x_2^N = 1000, x_3^N = 1500$, as it satisfies both individual and collective rationality, but it is not in the core. The coalition $\{1, 2\}$ has an excess of $\epsilon(\{1, 2\}, x^N) = (500 + 1000) - 1200 = 300$, so they could collectively pay 300 less by dropping agent 3 from their coalition.

The **marginal contribution** of an agent is $M_i = v(N) - v(N \setminus \{i\})$, meaning the change in the worth of the grand coalition when they join. The marginal contribution is the utopian allocation for the player, and provides a lower bound for what they can receive in the core. If they receive a better payoff as part of an imputation, it will not be in the core, as the rest of the coalition will be better off without them. In the previous example, $M_3 = 3000 - 1200 = 1800$, so any allocation where agent 3 pays less than 1800 will not be in the core, as agent 1 and 2 are better off without them.

2.1.2 Properties of core-selecting mechanisms

The purpose of this thesis will be to explore how to find the most fair allocation of costs and risk-associated cost. It is preferable for that allocation to be in the core, as that will indicate a stable allocation. A mechanism to select an allocation in the core will be called a **core-selecting mechanism** [11]. The choice of the core-selecting mechanism is central to how utility is distributed, as different mechanisms will yield different allocations. The mechanisms can be described with a series of properties.

Property 1: Efficiency: *The total social welfare of coalition is maximized, assuming every agent acts truthfully [25].*

If a mechanism does not yield the optimal decision, potential utility has been lost. By default, cooperative games yield an efficient solution, so this will be true for most mechanisms. A non-efficient mechanism will be one that leads to a decision that is not the best for the group, which requires the mechanism to interfere with the decision making process. Most mechanisms do not interfere with this, and select the decision that lead to the characteristic functions. In a prisoners dilemma context, it would be a mechanism that leads to one or more prisoners turning in each other.

Property 2: Incentive compatibility: *Every agent maximizes their own utility by acting truthfully.*

In order to ensure an efficient solution and to reduce the complexity of bidding for agents, it is important that agents are incentivized to reveal their types with the set of types for a player i being Θ_i . The types contain all the information the agents know about

themselves and other players. This can be the other players costs and utilities, even when not working together, and also the certainty of the player in this information.

If a core-selecting mechanism doesn't have this property, agents may misreport their preferences in order to obtain a better outcome. This is not only unfair, but can lead to making an inefficient decision and makes it more complicated for agents to report their preferences. This could be a prisoner lying about the size of their fines as part of another coalition, in order to obtain a better allocation or to convince the others that it is better to turn them in.

Two types of incentive compatibility will be considered: Dominant Strategy Incentive Compatibility (DSIC) and Bayesian Incentive Compatibility (BIC). DSIC means that revealing ones types is the optimal strategy, no matter the actions of the other agents, while BIC means that revealing ones types is optimal if all other agents tell their types [27].

Property 3: Budget balance: *The community manager does not have financial excess nor deficit.*

This property ensures that all generated utility is distributed among the agents. A deficit is detrimental for the community manager, as it will no longer be viable for them to maintain the community, and an excess is detrimental to the agents, as they do not receive all the utility they generate. This is distinct from efficiency, as efficiency regards the total welfare and budget balance regards the distributed welfare. If the prisoners have a mediator that handles how to distribute the fines, an efficient mechanism will yield the lowest possible total fine, while a budget balanced mechanism will make the prisoners pay exactly the fine. If the mechanism is not budget balanced, the mediator might take some payment for mediating, or pay part of the fine.

Property 4: Group rationality: *No group of agents can benefit from splitting from the community.* This property requires the mechanism to always return an allocation where every coalition has a non-positive excess. If this property does not hold, the long term cohesion of the grand coalition is at risk. A mechanism that is not group rational can still return a group rational allocation, but it is only group rational if it is always the case. Group rationality is distinct from collective rationality, and does not imply collective rationality nor stability. It is possibly for a group rational mechanism to return an allocation that is not in the core, if it is not budget balanced.

Property 5: Additivity: *The sum of allocations for a set of sub-games will be the allocation for the game created by adding together the sub-games*

If a mechanism is additive, the choice of how to define the game has less of an impact on the allocations. In a prisoners dilemma where the agents are sentenced twice, the overall game can be split into two sub-games, one for each sentencing. If a mechanism is additive, it will return the same total allocation whether each sentencing is handled as a separate game or the total sentence is handled at once. In the context of energy trading, this is relevant as each time period can be seen as an independent sub-game, so an additive mechanism will return the same allocation whether allocations are calculated on a hourly, daily, weekly or other basis.

The challenge in choosing a core-selection mechanism comes from it not being possible

for any mechanism to satisfy both efficiency, incentive compatibility, budget balance and group rationality at once [30]. Similarly, Green and Laffonts impossibility states that incentive compatible and efficient mechanisms cannot be budget balanced in the general quasi-linear domain [9], quasi-linearity meaning that the utility gained is the utility of the good they receive, subtracted what they paid for it. Therefore each mechanism has its own trade-off.

In sections 4 and 5, the problem and mechanisms will be studied from a theoretical point of view, before they are compared in section 7.

2.2 Optimization

A **linear program** is a linear objective function and a set of linear constraints that together form an optimization problem. The objective function is the first line, eq. 2.3 below, and will indicate whether the problem should be minimized or maximized, what the variables that must be changed are, and the coefficients of those variables. The following lines, eq. 2.4 and 2.5, will be the constraints that indicate the limitations on the variables.

$$\min_x cx \quad (2.3)$$

s.t.

$$ax \geq b \quad (2.4)$$

$$x \geq 0 \quad (2.5)$$

This linear program is deterministic, as it assumes 100% certainty in the form of the constraints. **Stochastic programming** provides a method to modeling uncertainties in the constraints or objectives. In this thesis, it will be by introducing a set of scenarios. In the stochastic example below, eq. 2.6 is the modified objective function. c_w can now take a different value depending on the scenario $w \in W$, and each scenario is weighted by their chance of happening π_w , causing the objective function to minimize the expected value. Scenarios can also be used in constraints, as eq. 2.7 is now defined for every scenario, so ax is constrained by the worst case.

x is not indexed by scenarios, therefore the value of x will be the same for all scenarios. A variable y_w would be able to take a different value for each scenario.

$$\min_x \sum_{w \in W} \pi_w c_w x \quad (2.6)$$

s.t.

$$ax \leq b_w \forall w \in W \quad (2.7)$$

$$x \geq 0 \quad (2.8)$$

CHAPTER 3

Modelling the problem

In order to model the power purchasing problem a three-step process is used as seen in figure 3.1. The primary focus for this thesis is the allocation mechanisms. The processes before are used to generate the characteristic functions, but can be replaced with other methods to fit the BRP.

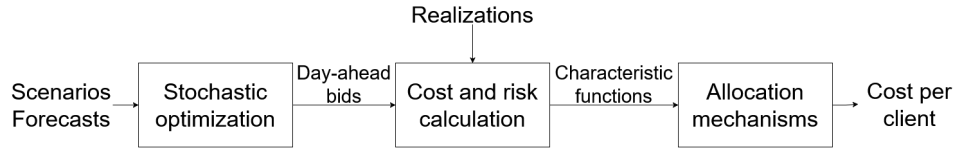


Figure 3.1: Model flowchart

3.1 Stochastic Optimization

For this implementation, a stochastic optimization is used to generate bids. The BRP can have many strategies in order to reduce their cost, they might not optimize, instead forecasting and bidding their forecast. In practice, they can be assumed to participate in more markets than just the day-ahead market, complicating this part of the model significantly. For this thesis, focus is kept on the day-ahead market.

The exact process the bids are generated doesn't make a difference for applying game theory, however, it is required that the process outputs a separate bid for each client, as game theory requires the knowledge of the worth of individual clients.

The model in equations 3.1 to 3.5 aims to minimize imbalance costs in a two-price scheme, is solved for a coalitions S , and the useful output will be the vector of day-ahead bids for the coalition $P_{t,S}^{DA}$

Variables:

$P_{t,S}^{DA}$ is the day ahead bid variable. It is sub-scripted by the time t , as the BRP gives a separate bid for each time period. It is also sub-scripted by S , the current coalition.

$\Delta_{w,t}$, $\Delta_{w,t}^+$, $\Delta_{w,t}^-$ are variables that indicate the imbalances the bids cause. $\Delta_{w,t}$ is the imbalance, while $\Delta_{w,t}^+$ and $\Delta_{w,t}^-$ are the positive and negative parts of the imbalance, respectively. These are sub-scripted by the time period t and the demand scenario w . As this is a stochastic model, the day ahead bid P_t^{DA} must be the one that minimizes the expected cost across all scenarios.

Objective function:

The model is given a set of scenarios W for the demand realization $P_{w,t,i}^D$ and scenarios

U for the dominant imbalance direction $\sigma_{u,t}$ and price $\lambda_{u,t}$. The dominant direction $\sigma_{u,t}$ is 1 if the dominant imbalance is positive and 0 if it is negative. Its inclusion in the objective function enables the two-price scheme, by only counting costs for imbalances in the dominant direction. π^λ and π^D indicate the weight of price scenarios $u \in U$ and demand scenarios $w \in W$ respectively. As all scenarios are assumed equiprobable, they are calculated $\pi^\lambda = \frac{1}{|U|}$ and $\pi^D = \frac{1}{|W|}$.

Constraints:

Equation 3.2 measures the deviation between demand $P_{w,t,i}^D$, production $P_{t,i}^G$, and day ahead bid $P_{t,S}^{DA}$. Production is given as a deterministic value based on a forecast for each client in this case. Demand and production are summed across all clients c in the coalition S .

Equation 3.3 separates the demand into positive and negative deviation, enabling differentiating between imbalance directions in the two price scheme.

$$\min_{P_{t,S}^{DA}, \Delta_{w,t}^+, \Delta_{w,t}^-} \sum_{u \in U} \sum_{w \in W} \pi^\lambda \pi^D \left(\sum_{t \in T} (\sigma_{u,t} \lambda_{u,t} \Delta_{w,t}^+ + (1 - \sigma_{u,t}) \lambda_{u,t} \Delta_{w,t}^-) \right) \quad (3.1)$$

s.t.

$$\Delta_{w,t} = \sum_{i \in S} P_{w,t,i}^D - \sum_{i \in S} P_{t,i}^G - P_{t,S}^{DA}, \quad \forall w \in W, t \in T \quad (3.2)$$

$$\Delta_{w,t}^+ - \Delta_{w,t}^- = \Delta_{w,t}, \quad \forall w \in W, t \in T \quad (3.3)$$

$$\Delta_{w,t}^+ \geq 0, \quad \forall w \in W, t \in T \quad (3.4)$$

$$\Delta_{w,t}^- \geq 0, \quad \forall w \in W, t \in T \quad (3.5)$$

Every constraint in this problem is defined for all $t \in T$, and therefore the problem is fully decomposable into smaller problems temporally. Multiple clients are only involved in the sums in constraint 3.2. As this constraint is a sum of the demands, productions and bids of individual clients, the optimal bid for a coalition will be the sum of singleton bids $P_{t,S}^{*DA} = \sum_{i \in S} P_{t,i}^{*DA}$, therefore it can be decomposed into singleton coalitions.

This singleton decomposed problem will be used instead of the aggregated optimization, as the imbalances of all combinations of clients can be generated from the bids of the singular clients. Adding together bids of individual clients is significantly faster than bespoke optimizations for each coalition.

3.2 Cost and risk calculation

The output of the optimization will be the optimal day ahead bidding decisions for each singleton coalition $P_{t,i}^{DA} \forall i \in N$.

These bids are compared with the realization of client demand and production ex-post, to obtain the real imbalance for each coalition.

$$\Delta_{t,i}^{Real} = P_{t,i}^{D,Real} - P_{t,i}^{G,Real} - P_{t,i}^{DA} \quad (3.6)$$

$$\Delta_{t,S}^{Real} = \sum_{i \in S} \Delta_{t,i}^{Real} \quad (3.7)$$

Clients pay the BRP the spot price for power, and therefore the imbalance spread is what is relevant to the BRP's cost. The spread will be the difference between the day ahead price and the imbalance price, $\lambda_t = \lambda_t^{Imbalance} - \lambda_t^{DA}$.

The imbalance cost to the BRP for a coalition for a given time will then be:

$$J_{t,S} = \lambda_t \Delta_{t,S}^{Real} \quad (3.8)$$

Two types of games will be studied in this thesis, the division of cost and the division of risk. Up until this point, the method to compute both of them is the same. The difference lies in what parameter is used to calculate the characteristic function.

3.2.1 Characteristic function for allocating costs

The game of allocating costs is simpler, and is viable in a market with an assymetric cost structure, like a two-price scheme or the upcoming full cost balancing model. The characteristic function for a coalition is simply the sum of their costs:

$$v(S) = \sum_{i \in S} \sum_{t \in T} J_{t,S} \quad (3.9)$$

In this thesis, imbalance costs $J_{t,S}$ are computed with a t for every 15 minutes, in line with the imbalance settlement period.

3.2.2 Characteristic function for assigning risk

The game of assigning the risk level of each client is slightly more complicated, as it requires defining a risk measure, but it is relevant for any market, not only one with assymetric pricing.

CVaR, also known as expected shortfall, is chosen as the risk metric. The purpose of the game is to minimize the CVaR of the costs in a time period. CVaR at confidence level α is defined as the average cost incurred in the worst $\alpha\%$ of imbalance intervals. Mathematically, this can be expressed as:

$$CVaR_{\alpha}(J_{t,S}) = \frac{1}{|\mathcal{T}_{\alpha}(J_{t,S})|} \sum_{t \in \mathcal{T}_{\alpha}(J_{t,S})} J_{t,S} \quad (3.10)$$

Where:

- $J_{t,S}$ is the set of imbalance costs in coalition S across all time periods.
- $\mathcal{T}_{\alpha}(J_{t,S})$ denotes the α -tail set of $J_{t,S}$, defined as: $\mathcal{T}_{\alpha}(J_{t,S}) = \{t \in T : J_{t,S} \leq VaR_{\alpha}(J_{t,S})\}$. i.e. it is the indexes t that have an income at or below VaR.
- $VaR_{\alpha}(J_{t,S})$ denotes the Value-at-Risk at confidence level α , representing the cost threshold below which $(1 - \alpha) \cdot 100\%$ of the outcomes lie. In other words, it identifies the minimum cost such that only the worst $\alpha\%$ of time steps exceed it.

For this thesis, the empirical VaR is used, computed directly from the discrete set of time step outcomes $J_{t,S}$. Formally, it is defined as: $\text{VaR}_\alpha(J_{t,S}) = \inf \{ \zeta \in \mathbb{R} \mid \mathbb{P}(J_{t,S} \leq \zeta) \geq 1 - \alpha \}$. In simpler terms, $\text{CVaR}_\alpha(J_{t,S})$ measures the average imbalance cost during the worst-case α -percentile periods.

For this thesis $\alpha = 0.05$, corresponding to the 5% worst case scenarios.

The characteristic function of the game is then:

$$v(S) = \text{CVaR}_\alpha(J_{t,S}) \tag{3.11}$$

CHAPTER 4

Game Theoretical Details of the Problems

In this section, the specific properties of the games that arise from the cost allocation and risk assigning problems will be studied. These properties help describe the problem and are useful for describing the properties of the core-selecting mechanisms in section 5.

4.1 Incentive compatibility

The purpose of incentive compatibility is to get the agents to reveal their types Θ_i . Incentive compatibility will not get the agents to only reveal their demand accurately. In game theory, the type includes all the information the agent knows that is relevant to the game, and as such the types for this game will include their future demand, knowledge about someone else's future demand, information about future prices or dominant directions, and the agents certainty in all of these.

An incentive compatible mechanism will encourage an agent to reveal what they expect their own demand will be, and publicly reveal any information they have about other agents demand or price information.

This will make an agent reveal their types Θ_i as "I expect to have a demand of x , and everyone else will have an imbalance of y " when joining a coalition, which will reduce the imbalance of the coalition when the agent joins it. If a mechanism is not incentive compatible, the agent might strategically reveal their type as "I expect to have a demand of $x-y$ ". The problem with such a mechanism, is if two separate agents have the same information, they might both say "I expect to have a demand of $x-y$ ", causing an imbalance correction of $2y$, which would be an overcorrection.

An example of a Dominant Strategy Incentive Compatible mechanism is VCG[32]. VCG is DSIC because agents get allocations based on the value they add to all other agents. More specific information on VCG in section 5.4. This will encourage the agent to say "I expect to have a demand of x , and everyone else will have an imbalance of y ", because the agent is not paid based on their own imbalance, instead they are paid for the value they add to the coalition, including the value they add from revealing their types.

4.2 The cost allocation problem

When dividing costs for the imbalance reduction problem, each instance where costs are divided can be seen as a separate game. The amount of realizations of imbalance in each game is a choice that must be made when dividing costs. A realization of imbalance will

be when the difference between the traded electricity and the actual demand is measured. For this thesis, a realization of imbalance t will be 15 minutes, in line with the imbalance settlement period in the nordic countries [26].

For this section, the games that result from a single hour will be studied. These games are relevant if deciding on an internal imbalance settlement period in the portfolio of 15 minutes. They can also be used to analyze how games look for longer periods. As a game of 24 hours consists of 96 15-minute subgames, the subgames have an effect, especially if the mechanism is additive.

The games will be cost reduction games, and therefore will be minimization problems.

4.2.1 Possible game structures

For every realization of imbalance, the problem can result in one of 4 structures of games, based on the imbalances in relation to each other.

- **Extreme imbalance:** All coalitions have the same imbalance sign.
- **High imbalance, dominant direction:** All agents with the same sign as the net imbalance has a smaller imbalance than the net imbalance, and the overall imbalance of the portfolio is in the dominant direction of the power market. Results in no agent being able to change the sign of the net imbalance, by leaving the grand coalition.
- **Non-dominant direction imbalance:** The overall imbalance is opposite the market imbalance direction.
- **Low imbalance:** If none of the other cases are true.

4.2.2 Sub-additivity and convexity

Sub-additivity is the minimization version of superadditivity, and holds if:

$$v(S_1 \cup S_2) \leq v(S_1) + v(S_2), \quad \forall S_1, S_2 \in N \text{ such that } S_1 \cap S_2 = \emptyset \quad (4.1)$$

Meaning that the costs of a two coalitions together must be lower than or equal to their costs apart. This is true for the cost problem.

Proof. Sub-additivity of the cost problem can be proven by considering the imbalance $\Delta_{t,S}^{\text{Real}}$. As $v(S_1 \cup S_2) = \max(0, \lambda_t(\Delta_{t,S_1}^{\text{Real}} + \Delta_{t,S_2}^{\text{Real}}))$, it is clear that the cost is highest when the imbalances have the same sign, as in that case $v(S_1 \cup S_2) = v(S_1) + v(S_2)$ if $\Delta_{t,S_1}^{\text{Real}} \cdot \Delta_{t,S_2}^{\text{Real}} \geq 0$. If they have the opposite sign, $v(S_1 \cup S_2) < v(S_1) + v(S_2)$ if $\Delta_{t,S_1}^{\text{Real}} \cdot \Delta_{t,S_2}^{\text{Real}} < 0$, as the imbalance of one will counter the other. Therefore the problem for every imbalance realization is sub-additive.

As every imbalance realization is an independent sub-game, the game v is the sum of two games u, w such that $v(S) = u(S) + w(S)$. As the games are independent,

if sub-additivity holds for both sub-games u and w , it will also hold for the sum of them v .

Recall that concavity is a stronger condition than sub-additivity, and holds if the following is true:

$$v(S_1 \cup S_2) + v(S_1 \cap S_2) \leq v(S_1) + v(S_2) \forall S_1, S_2 \subseteq N \quad (4.2)$$

I.e. the sum of the cost of two coalitions should be greater or equal to the sum of the cost of the union + the intersection of them. This means that the marginal contribution of a player to a coalition should be lower than their marginal contribution to any subcoalition. Concavity does not hold for the cost problem.

Proof. Concavity can be disproven by counterexample. Given a game with three agents where $\Delta_1^{\text{Real}} = \Delta_2^{\text{Real}} = -\Delta_3^{\text{Real}} = 1kWh$ with a positive dominant direction and an imbalance spread of $1\frac{EUR}{kWh}$, the coalitions $S_1 = \{1, 3\}$ and $S_2 = \{2, 3\}$ will have a cost of $v(S_1) = v(S_2) = 0EUR$, but $v(N) + v(\{3\}) = 1EUR + 0EUR$. Since $v(N) + v(\{3\}) > v(S_1) + v(S_2)$ concavity does not hold for this game.

4.2.3 Existence of a core

The core will be the set of imputations where all coalitions have a non-positive excess with respect to the grand coalition. In the cost game, the core is always nonempty, and the set of imputations in the core is greatly affected by the imbalance conditions in the portfolio. In this section, this will be proven and the conditions that form the core will be studied.

While the existence of the core can be proven with the Bondareva-Shapley theorem [35] as is done in section 4.3.2 for the risk problem, here, the core will be found separately for each imbalance case to understand how the core behaves.

Note that these are cases for a single imbalance realization t . This analysis is especially relevant to the additive mechanisms and to mechanisms where the allocation is calculated based on individual time steps.

Extreme imbalance

The simplest is the case of *extreme imbalance*. This will be an inessential game where the allocation $x_i^N = v(\{i\})$ is the unique imputation in the core.

Proof. For an allocation to be an imputation, individual rationality and collective rationality must hold. For individual rationality to hold, each agent must pay no more than their singleton cost. For collective rationality to hold, the sum of payments must equal the total costs.

In the case of extreme imbalance, the imbalances will always be in the same direction. In the cost game, if imbalances are in the same direction, then $v(S_1) + v(S_2) = v(S_1 + S_2) \forall S_1, S_2 \subseteq N$, meaning that the game is inessential.

Because the game is inessential, the allocation where every agent pays their singleton

cost $x_i^N = v(\{i\})$ will be the only individually rational allocation. This allocation will be collectively rational, as $\sum_{i \in N} v(\{i\}) = v(N)$ for an inessential game. It will also be collectively rational, as $\sum_{i \in S} v(\{i\}) = v(S)$, so no coalition can get a better payoff. Therefore it will be the unique imputation in the core.

High imbalance, dominant direction

For the case of *high imbalance, dominant direction*, the core will contain a single imputation that is: $x_i = \sum_{t \in T} \lambda_t \cdot \Delta_{i,t}^{\text{Real}}$

Proof. For this game, agents can be divided into imbalance causers and imbalance reducers. The imbalance causers will be the agents whose marginal contribution is their imbalance multiplied with the imbalance spread, if the cost is positive (equivalent to their singleton costs):

$$\delta^+ = \{i \in N \mid M_i = v(\{i\})\} \Leftrightarrow \delta^+ = \{i \in N \mid M_i = \lambda_t \cdot \Delta_{i,t}^{\text{Real}} \wedge M_i \geq 0\} \quad (4.3)$$

And the imbalance reducers will be those whose marginal contribution is their imbalance multiplied with the imbalance spread, if the cost is negative:

$$\delta^- = \{i \in N \mid M_i = \lambda_t \cdot \Delta_{i,t}^{\text{Real}} \wedge M_i < 0\} \quad (4.4)$$

The sets are interesting because their marginal contribution is a linear function of their imbalance. These exist for the other cases as well, but the high imbalance case is defined by the following:

$$N = \delta^+ \cup \delta^- \quad (4.5)$$

Additivity of marginal contributions means that the sum of marginal contributions of clients in a coalition sum up to the characteristic function of the coalition. Additivity of marginal contributions at the top level, means that this holds for the grand coalition:

$$\sum_{i \in N} (M_i) = v(N) \quad (4.6)$$

As $v_t(N) = \sum_{i \in N} \Delta_{i,t}^{\text{Real}} \cdot \lambda_t = \sum_{i \in N} M_i$, additivity of marginal contributions hold at the top level.

For an imputation to be in the core, every coalition must have a non-positive excess with respect to the grand coalition[25]. The imbalance causers are characterized with $M_i = v(\{i\}) \forall i \in \delta^+$. For $\epsilon(S, x^N) \leq 0$ to hold for all coalitions, they must pay at least their utopian allocation. Inserting the marginal contribution of δ^+ gives:

$$x_i^N \geq M_i \Leftrightarrow x_i^N \geq v(\{i\}) \forall i \in \delta^+ \quad (4.7)$$

Taking into account that individual rationality requires $x_i^S \leq v(\{i\})$ results in only a single stable payoff for imbalance causers, given in equation 4.8. This argument

holds in general across game structures, and shows that all clients in δ^+ must pay their singleton costs to be in the core.

$$x_i^N = v(\{i\}) = M_i \forall i \in \delta^+ \quad (4.8)$$

For collective rationality to hold, the sum of payments must equal the total cost. If $N = \delta^+ \cup \delta^-$ as it is in the high imbalance case, then the following is required for collective rationality to hold:

$$\sum_{i \in \delta^-} (x_i) + \sum_{j \in \delta^+} (x_j) = v(N) \quad (4.9)$$

Using the result from eq. 4.8 gives:

$$\sum_{i \in \delta^-} (x_i) + \sum_{j \in \delta^+} (M_j) = v(N) \quad (4.10)$$

Using equation 4.6 and 4.5 leads to

$$\sum_{i \in \delta^-} (x_i) + \sum_{j \in \delta^+} (M_j) = \sum_{i \in \delta^-} (M_i) + \sum_{j \in \delta^+} (M_j) \Leftrightarrow \sum_{i \in \delta^-} (x_i) = \sum_{i \in \delta^-} (M_i) \quad (4.11)$$

As the marginal contribution is an upper bound for the allocation to imbalance reducers, no one can pay less than their marginal contribution. Therefore the only collectively rational allocation is:

$$x_i^N = M_i = \lambda_t \cdot \Delta_{i,t}^{\text{Real}} \forall i \in \delta^+ \quad (4.12)$$

As such, the core will contain a unique imputation in the high imbalance case. This imputation will be called the *Uniform Price mechanism* for this thesis, and is the allocation where each agent is paid their imbalance multiplied with the imbalance spread. For all cases except the *Non-dominant direction imbalance* it can be defined: $x_i = \sum_{t \in T} \lambda_t \cdot \Delta_{i,t}^{\text{Real}}$.

Low imbalance

In the case of *low imbalance* the core will be non-empty and still contain the allocation $x_i = \sum_{t \in T} \lambda_t \cdot \Delta_{i,t}^{\text{Real}}$.

Proof. The difference between *high imbalance*, *dominant direction* and *low imbalance* is that this case contains a new set δ' that encapsulates the clients that are not strictly marginal causers or reducers, these clients will have an imbalance of the same sign as the net imbalance, but with a higher magnitude than the net imbalance, they are defined by their marginal contribution:

$$0 \leq M_i < v(\{i\}) \forall i \in \delta' \quad (4.13)$$

The addition of this set changes the makeup of N :

$$N = \delta^+ \cup \delta^- \cup \delta' \quad (4.14)$$

Payments for δ^+ can still be found by eq. 4.8.

Payments for δ^- must still be worse than the utopian allocation, but introducing the new set δ' causes additivity of marginal contributions at the top level to no longer hold, and means that imbalance reducers no longer share the entire overpayment of imbalance causers. In this case, it is possible to have an imputation in the core where $\sum_{i \in \delta^-} (x_i) > \sum_{i \in \delta^-} (M_i)$ and $\sum_{i \in \delta'} (x_i) < \sum_{i \in \delta'} (v(\{i\}))$.

The Uniform Price allocation will still be in the core for the low imbalance case. It remains both individually and collectively rational in this case, so it is an imputation. Since all imbalance reducers get paid this utopian allocation, there will be no coalition that is more preferable to them, and since a game without imbalance reducers is inessential, it is not possible to make a better coalition without them. The Uniform Price allocation will be in the core. But it will be at the very edge of the core, since the excess of agents in δ^+ and δ' with respect to their singleton coalition is 0.

Non-dominant direction imbalance:

In this case, the core will be non-empty but will not contain the allocation $x_i = \sum_{t \in T} \lambda_t \cdot \Delta_{i,t}^{\text{Real}}$.

Proof. When the overall imbalance of the portfolio is opposite of the market, the characteristic function will be $v(N) = 0$.

The lowest possible cost is $v(S) = 0$ both for multi-client and singleton coalitions, so the allocation $x_i^N = 0 \forall i \in N$ will always be an imputation, as it satisfies both individual and collective rationality. It will also be in the core, since no coalition can have a $v(S) < 0$, so no coalition can have a positive excess. Therefore the core is guaranteed to exist.

In this case, the payments from clients with imbalances in the market dominant direction will not be able to cover paying clients with imbalances in the other direction the imbalance price because: $\sum_{i \in N; v(\{i\}) > 0} \lambda_t \cdot \Delta_{t,i}^{\text{Real}} < \sum_{i \in N; v(\{i\}) = 0} \lambda_t \cdot \Delta_{t,i}^{\text{Real}}$.

This causes the Uniform Price allocation to not be budget balanced in this case. Modifications that enable budget balance are discussed in the section dedicated to Uniform Price, section 5.2.

4.2.4 Overall stability of the game

As explained in this section, the core of the game is tightly constrained if many clients are in δ^+ . But these studied cases are made on the basis of individual hours. If costs are divided over 24 hours or longer, the core can become much wider as clients are unlikely to be in δ^+ for every time t .

In general, the core is smaller when there are many agents, the agents are comparable in

size, and there is a high imbalance in a single direction, as this increases the chance of an extreme- and high imbalance result. The difficulty lies in distributing gains when the core is large.

4.3 The risk assigning problem

When assigning contribution to risk, the problem is similar to the cost problem but has some different properties. When using CVaR as a metric, games can no longer easily be broken into subgames for every imbalance realization, as the CVaR is a statistic depending on the whole period.

4.3.1 Sub-additivity and convexity

Sub-additivity holds if:

$$v(S_1 \cup S_2) \leq v(S_1) + v(S_2), \quad \forall S_1, S_2 \in N \text{ such that } S_1 \cap S_2 = \emptyset \quad (4.15)$$

Sub-additivity will hold for the risk problem as well.

Proof. the set $\mathcal{T}_\alpha(J_{t,S})$ is the set of indexes for the α highest values of $J_{t,S}$, then the maximal value of $v(S_1) + v(S_2)$ will be found when they have the same worst case scenarios, in which case that will also be the worst case scenarios for $S_1 \cup S_2$: $\mathcal{T}_\alpha(J_{t,S_1}) = \mathcal{T}_\alpha(J_{t,S_2}) = \mathcal{T}_\alpha(J_{t,S_1 \cup S_2})$. In that case, $v(S_1 \cup S_2) = v(S_1) + v(S_2)$, and in all other cases $v(S_1 \cup S_2) < v(S_1) + v(S_2)$, meaning that sub-additivity holds.

Concavity does not hold for the risk problem either.

Proof. Concavity can be disproven by counterexample for this game. Given a game with three agents A, B, and C: $\alpha = 0.5$, imbalances $\Delta_{t,\{A\}}^{Real} = \Delta_{t,\{B\}}^{Real} = -\Delta_{t,\{C\}}^{Real} = [1, 0]$, the coalitions $\{A, C\}$ and $\{B, C\}$ will have a CVaR of $v(\{A, C\}) = v(\{B, C\}) = 0$, but $v(N) + v(\{C\}) = 1 + 0$. As $v(N) + v(\{C\}) > v(\{A, C\}) + v(\{B, C\})$, concavity does not hold for this game.

4.3.2 Existence of a core

All risk games will be balanced and have a non-empty core.

Proof. The existence of a core in all cases can be proven by using the Bondareva-Shapley theorem[35]. The theorem says that a game has a non-empty core if and only if the game is balanced. A game is balanced if for every balancing function $\beta : 2^N \setminus \{\emptyset\} \rightarrow [0, 1]$ where:

$$\sum_{S \in 2^N : i \in S} \beta(S) = 1 \quad (4.16)$$

It holds that (adapted to being a cost reduction game):

$$\sum_{S \in 2^N \setminus \{\emptyset\}} (\beta(S)v(S)) \geq v(N) \quad (4.17)$$

In other words, a game is balanced if it is not possible to make a set of subcoalitions that have a lower cost than the grand coalition, even if agents are allowed to partially enter coalitions. Intuitively, this is true when using CVaR as a metric, as CVaR will pick the worst case scenarios. Analytically, it can be shown by applying the theorem. To apply the theorem, first it is used that the CVaR function is convex, such that the CVaR of the sum of coalitions will at most be the sum of CVaR of coalitions:

$$\sum_{S \in 2^N} \beta(S) CVaR_\alpha(J_{t,S}) \geq CVaR_\alpha \left(\sum_{S \in 2^N} \beta(S) J_{t,S} \right) \quad (4.18)$$

The left hand side is equivalent to the sum of characteristic functions:

$$\sum_{S \in 2^N} \beta(S) CVaR_\alpha(J_{t,S}) = \sum_{S \in 2^N} \beta(S) v(S) \quad (4.19)$$

Using eq. 4.16, the balancing function must sum up to 1 for every agent. Because $J_{t,S} = \sum_{i \in S} J_{t,i}$, then the following is true:

$$CVaR_\alpha \left(\sum_{S \in 2^N} \beta(S) J_{t,S} \right) = CVaR_\alpha(J_{t,N}) = v(N) \quad (4.20)$$

Inserting into equation 4.18:

$$\sum_{S \in 2^N} \beta(S) v(S) \geq v(N) \quad (4.21)$$

As such, any risk game will be balanced and the core will always be non-empty.

CHAPTER 5

Core-selection mechanisms

In this section, the implemented core selection mechanisms will be described in detail. This includes both a general introduction to the mechanism, and how it works for the cost and risk problems.

To give some intuition on the mechanisms, a series of figures, like fig. 5.1, will show how the mechanisms allocate costs. These include both the two-price and one-price costs. The one-price costs are not used to calculate the allocation, but are there to show the imbalance of client C and show how the cost problem would look in a one-price scheme.

5.1 Shapley Value

A Shapley Value is a value function $\Phi_i(v)$ that assigns payoffs from the game v . Note that the game v is distinct from the characteristic function of a coalition in the game $v(S)$.

Shapley's axioms help define the Shapley Value:

- **Symmetry:** If agents i and j are substitutes, meaning that any coalition without them gains the same amount of utility when either of them are added, then their payoff should be the same, $\Phi_i(v) = \Phi_j(v)$
- **Null agent condition:** If i is a null agent, meaning that the utility of all coalitions without them does not change if they are added, their payoff should be 0, $\Phi_i(v) = 0$
- **Efficiency:** The sum of payoffs must equal the characteristic function for the coalition, $\sum_{i \in N} (\Phi_i(v)) = v(N)$
- **Additivity:** For any two games v, w played by the same agents. The payoff for the two games played together should equal the sum of the payoffs for the games for all agents $\Phi_i(v + w) = \Phi_i(v) + \Phi_i(w)$.

The unique value satisfying these axioms is called the Shapley Value [6]. The Shapley Value is a representation of the expected pay-off from participation in a game. Each agent in a game will have a marginal contribution to each possible coalition, meaning the value they add to that coalition. The Shapley Value is a mathematical expectation where it is assumed that the grand coalition will be built up by adding agents in a random order, and is calculated by equation 5.1. The fraction being the number of permutations of the specific coalition S , multiplied by the number of permutations agents can join the coalition after i , divided by the total coalitions. This is multiplied by the marginal

contribution of i to the specific coalition.

$$\Phi_i(v) = x_i^N = \sum_{S \in 2^N \setminus \{i\}} \frac{(|S|)! (|N| - |S| - 1)!}{N!} (v(S \cup \{i\}) - v(S)) \quad (5.1)$$

Allocating according to the Shapley Value means calculating a weighted sum of the difference between the value of each coalition containing i and the same coalition without i . This necessitates having the characteristic function of every possible coalition, meaning that the optimization must be done $2^{|N|}$ times, making it difficult to scale to larger problems.

The efficiency axiom of the Shapley Value ensures both the efficiency and budget balance properties from section 2.1.2.

Incentive Compatibility

While the Shapley Value is not generally incentive compatible, it is if the agents reveal their true costs and utilities [40]. In these problems, it is assumed that this is known ex-post, and the Shapley Value is therefore incentive compatible in this case.

Group Rationality

The Shapley Value is not guaranteed to be part of the core, unless the problem is convex (or concave for a minimization game like this). As was shown in section 4.2.2, the problem is not concave, and Shapley is therefore not guaranteed to be in the core. Distributing with basis in the Shapley Value does therefore not fulfill the group rationality property from section 2.1.2 in this case. This can lead to instability in the long run, as some shareholders may be displeased with remaining in the grand coalition.

In figure 5.1, an example of how Shaply allocates costs can be seen. The cross lined column indicates the singleton cost in a one-price scheme. Allocations are based on the two-price scheme, but one-price is included to visualize the imbalance of client C. For simplicity, an imbalance spread of $1 \frac{EUR}{kWh}$ with a positive dominant direction is used for this plot. Client A has a positive imbalance of $1kWh$, client B has a positive imbalance of $2kWh$ and client C has a negative imbalance of $2kWh$. The singleton imbalance costs indicate the costs for a non-aggregated client. Client A and B pay their full imbalance, while C get nothing because of the two-price system.

When applying the Shapley Value to this, A and B get a discount, with B getting a larger discount than A. C now gets paid for their reduction in imbalance cost. Even though their imbalance is the opposite of client B, they get paid slightly less. This comes as a result of client C not being able to capitalize on their non-dominant imbalance direction when they are in a singleton coalition. The allocation is budget balanced, as costs sum up to 1€.

5.2 Uniform Price

Uniform Price as implemented in this thesis is a way to allocate costs based on the marginal contribution of each client. The basic concept is that each client that causes a cost, pays their singleton costs, and each client who reduces costs shares the difference

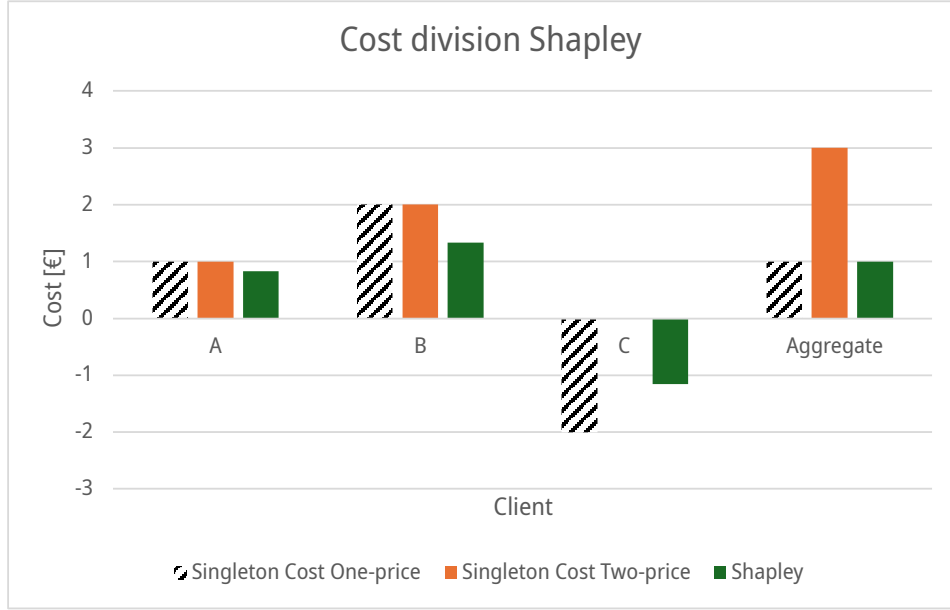


Figure 5.1: Cost allocation for Shapley mechanism

between the payments of the cost causers and the total cost. The implementation is slightly different for each game, so they are described separately.

5.2.1 Uniform Price applied to cost allocation

In this section, the application of Uniform Price specifically to the cost allocation problem will be studied. It is formulated as follows:

$$x_i = \sum_{t \in T} \begin{cases} -k_t^{\text{uni}} \cdot |\Delta_{t,i}^{\text{Real}}|, & \text{if } v(\{i\}) = 0. \\ v_t(\{i\}), & \text{otherwise.} \end{cases} \quad (5.2)$$

With k_t^{uni} being how much dominant direction imbalance each unit of non-dominant direction imbalance countered, and v_t being the subgame that is the imbalance realization at time t .

k_t^{uni} is only needed in an assymetric price scheme and is calculated in equation 5.3. $k_t^{\text{uni}} = 1$ for all cases from section 4.2.3 except *non-dominant direction imbalance*. The denominator gives the difference between the costs of the singleton coalitions and the cost of the grand coalition. This is the "overpayment" of the cost adding agents for that t . This is divided by the total imbalance of the clients in the non-dominant direction, to obtain how much cost each unit of imbalance in the non-dominant direction reduced the total cost with. k_t^{uni} will be 1 when the overall imbalance direction in the portfolio matches the dominant imbalance direction in the imbalance market, and will in the range $0 \leq k_t^{\text{uni}} < 1$ when the imbalance direction of the portfolio and the market are opposite.

$$k_t^{\text{uni}} = \frac{\sum_{i \in N, v_t(\{i\}) > 0} v_t(\{i\}) - v_t(N)}{\sum_{i \in N, v_t(\{i\}) = 0} \Delta_{t,i}^{\text{Real}}} \quad (5.3)$$

The logic is that if an agent has an imbalance in the dominant imbalance direction, they pay their singleton cost. If an agent has an imbalance in the opposite direction, they are compensated by the cost of the imbalance they countered.

Because the Uniform Price allocation is not budget balanced in the *non-dominant direction imbalance* case, as shown in section 4.2.3, there are three possibilities for what to do in this case:

1. **Unmodified:** Applying the Uniform Price allocation unmodified will not be budget balanced, as payments to cost reducers will be higher than payments from cost causers
2. k_t^{uni} : This will reduce the payments to cost reducers in this case, resulting in the allocation not being in the core.
3. **0 Payments:** Returning an allocation where everyone pays 0 if $v(N) = 0$ will keep the mechanism budget balanced and group rational, but will make it inconsistent as cost causers will no longer be punished in this case, and make it no longer continuous.

k_t^{uni} is chosen to keep the allocation continuous and consistent across cases, in order to preserve the central idea of the allocation: that cost causers must pay the full singleton cost of their imbalance.

Group Rationality:

The Uniform Price mechanism applied to the cost problem will not be group rational when using the k_t^{uni} modification.

Proof. Group rationality of the Uniform Price allocation can be disproven by counterexample: Suppose $\Delta_{1,A}^{\text{Real}} = 1, \Delta_{1,B}^{\text{Real}} = -1, \Delta_{1,C}^{\text{Real}} = -1, \lambda_1 = 1$ and positive is the dominant direction. Using k_t^{uni} , the allocation will be $x_A = 1, x_B = -0.5, x_C = -0.5$. This allocation is not in the core, as $v(\{A, B\}) = 0$ and $x_A^N + x_B^N = 0.5$, so A and B can gain by splitting from C.

Incentive compatibility:

The Uniform Price allocation applied to cost is not incentive compatible. Because the allocation rewards agents opposite the dominant direction directly from their own imbalance $\Delta_{t,i}^{\text{Real}}$, the agents are incentivized to keep their knowledge secret if they can predict the dominant direction and apply it to their own imbalance. So instead of revealing their type correctly, they are likely to bake it into their personal demand bid, to get an imbalance that is beneficial to them.

Proof. Incentive compatibility can be disproven by counterexample: Three clients each expect a demand of 1. Client A predicts that client B will have a true demand of 2, and predicts that the dominant direction in the power system will be positive.

This mechanism incentivizes client A to bid 0 in this case, as then they will get paid $k_t^{\text{uni}} \cdot 1$, instead of revealing their types.

An issue with this allocation is that it is very one-sided. Everyone on one side pays their full cost, while the other side gets paid. It is always at the edge of the core, with many coalitions having an excess of 0. The paying agents might ask why they should pay the helping agents, since the payers get no benefit from being in the coalition. An agent in the set δ' under this allocation has nothing to gain from staying, but can cause harm to the rest of the coalition by leaving. The severity of this issue is lessened for two reasons:

From a cooperative game perspective, it is assumed that players can make long term contracts and only focus on maximizing their own profit. Players are still in the core, even if they make the same amount of money as they would alone, so they help others even if it doesn't help themselves.

Since imbalance situation happens every 15 minutes, agents are not disadvantaged from being in δ' forever.

In figure 5.2, the Uniform Price cost example is given. Note that when the dominant direction in the portfolio is the same as in the market, this is equivalent to making an internal one-price system. Polluters pay their full imbalance, while non-dominant imbalance direction clients get paid their full imbalance. If the dominant imbalance direction of the portfolio is not the same as the overall system.

In figure 5.3, the allocation when the dominant direction of the portfolio and system is not the same is shown. Client C now has a negative imbalance of 4, causing an overall negative imbalance. In the two price scheme, this causes a cost of 0. Clients A and B still pay their singleton costs, but now client C only receives the sum of client A and B payments because $k_t^{\text{uni}} = \frac{3}{4}$.

5.2.2 Uniform Price applied to risk

Similar methods can be applied to the risk problem. This is done by looking at the costs each client adds to the α -tail set, and is computed as follows:

$$x_i = \frac{1}{|\mathcal{T}_\alpha(J_{t,N})|} \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} \begin{cases} -k_t^{\text{uni}} \cdot |\Delta_{t,i}^{\text{Real}}|, & \text{if } J_{t,i} = 0. \\ J_{t,i}, & \text{otherwise.} \end{cases} \quad (5.4)$$

With k_t^{uni} being the same as defined in equation 5.3. For risk, it will almost always be 1, as it is only less if there is profit in one of the α worst scenarios. The method is very similar as the implementation for cost, except only the time periods in the worst case set $t \in \mathcal{T}_\alpha(J_{t,N})$ are considered, and it is divided by the amount of time periods in the worst case set.

Group rationality

The allocations produced by the Uniform Price mechanism applied to risk will be group rational and always in the core.

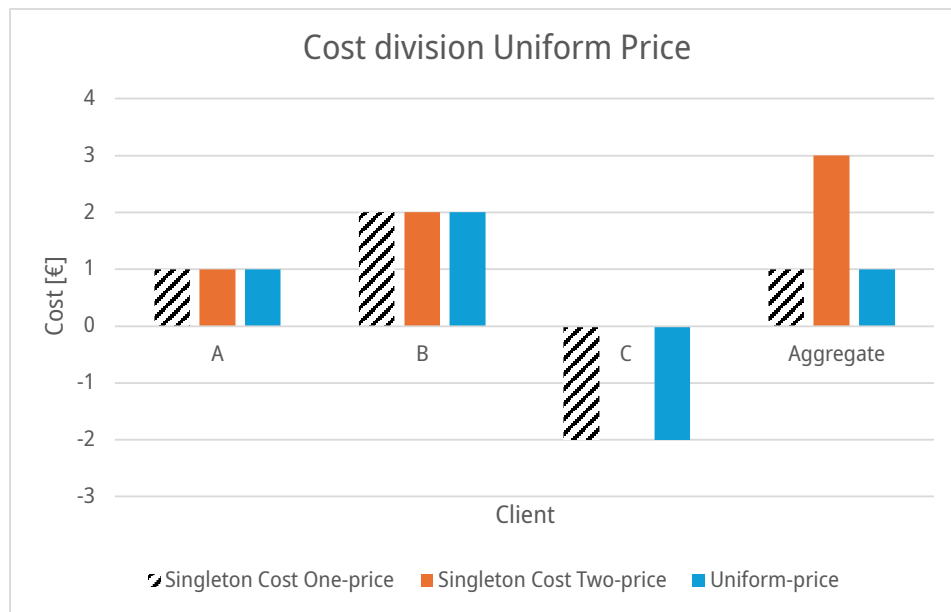


Figure 5.2: Cost allocation for Uniform Price mechanism

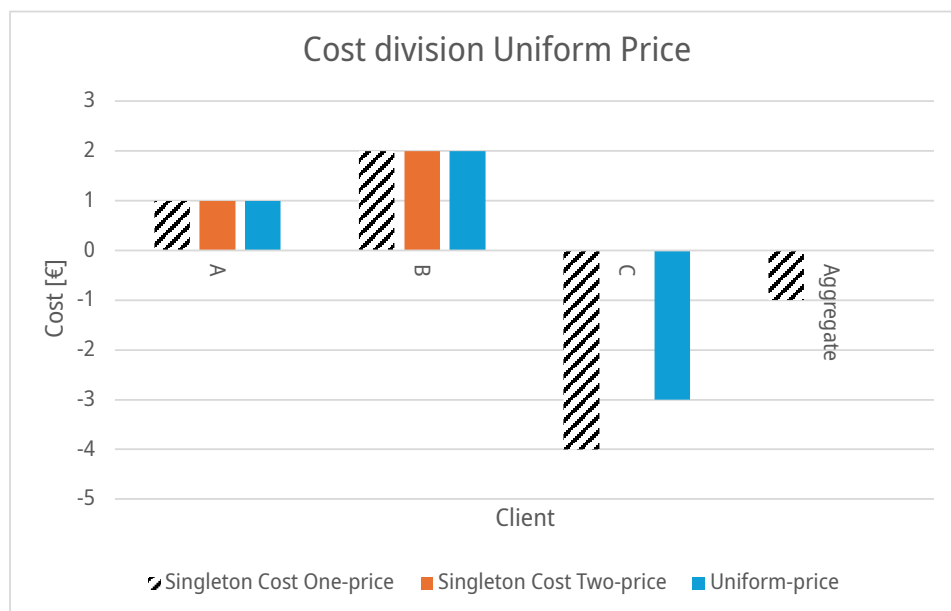


Figure 5.3: Cost allocation for Uniform Price mechanism with dominant direction of portfolio and system opposite

Proof. For an allocation in the core, it must be individually rational, so $v\{i\} \geq x_i(N) \forall i \in N$. In this game, it is not possible for $v\{i\} < x_i(N)$ as it is equivalent to $\sum_{t \in \mathcal{T}_\alpha(J_{t,\{i\}})} J_{t,\{i\}} < \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} J_{t,\{i\}}$. As the set $\mathcal{T}_\alpha(J_{t,\{i\}})$ will be the one that maximizes $\sum_{t \in \mathcal{T}_\alpha(J_{t,\{i\}})} J_{t,\{i\}}$, this is never the case.

Another condition is collective rationality, that $\sum_{i \in N} x_i(N) = v(N)$. This is ensured, as $\sum_{i \in N} \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} J_{t,\{i\}} = v(N)$.

Finally, the excess must be non positive with respect to all subcoalitions. The proof here is similar to individual rationality. For group rationality to not hold, the following must be true: $\exists S \subset N$ such that $v\{S\} < \sum_{i \in S} x_i(N)$. This is equivalent to $\sum_{t \in \mathcal{T}_\alpha(J_{t,S})} J_{t,S} < \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} J_{t,S}$, but since the set $\mathcal{T}_\alpha(J_{t,S})$ will be the one that maximizes $\sum_{t \in \mathcal{T}_\alpha(J_{t,S})} J_{t,S}$, this is never the case.

Therefore, the allocation will always be in the core.

Incentive-compatibility

The mechanism is not incentive compatible for the same reasons as the Uniform Price applied to the cost problem. Since agents are rewarded partly from their own imbalance, they can bid their demand strategically.

5.3 Reduced Cost

Reduced Cost will in this thesis be defined as follows and only implemented for the cost problem:

$$x_i^N = \sum_{t \in T} \frac{v_t(\{i\})}{\sum_{j \in N} v_t(\{j\})} \cdot v_t(N) \quad (5.5)$$

Where v_t is the subgame resulting from the imbalance realization period t . The logic from this allocation is that agents with an imbalance in the dominant direction pay their share of the total cost according to imbalance size, while agents with imbalances in the other direction pay nothing and do not get paid. In this way, it removes some of the incentive from strategic bidding, while keeping the allocation budget balanced. This allocation is presented by the Danish TSO Energinet for netting in the full cost balancing scheme [15].

This is the opposite extreme of the Uniform Price mechanism. With Uniform Price, the agents with dominant direction imbalances pay their singleton costs, which is the worst possible outcome for them that is individually rational, while the non-dominant direction agents receive all the benefit from the aggregation. In the Reduced Cost mechanism, the non-dominant direction agents get their singleton payout, while dominant direction agents get all the gains from aggregation.

Group rationality

The allocation produced by the Reduced Cost mechanism is not always group rational, as it will be possible for agents to group up for lower costs if a coalition has a lower imbalance than the grand coalition.

Proof. Group rationality can be disproven by counterexample: $\Delta_{1,A}^{Real} = 1, \Delta_{1,B}^{Real} = 1, \Delta_{1,C}^{Real} = -1, \lambda_1 = 1$ and positive is the dominant direction. The Reduced Cost allocation will be $x_A = 0.5, x_B = 0.5, x_C = -0.5$. Since $v(\{A, C\}) = 0 < x_A^N + x_C^N = 0.5$, the allocation is not in the core.

Incentive compatibility:

This mechanism is DSIC if the true costs for the agents are known.

Proof. As the allocation is based on the singleton cost of an agent. If true costs are known, an agent cannot lie about their singleton cost $v(\{i\})$. If an agent has information they believe could reduce the costs of other agents, they can choose to reveal it. If revealing their type Θ_i reduces the costs of one or more agents $v(\{j\})$ by a vector of nonnegative numbers y_t and the grand coalition cost $v(N)$ by $0 \leq z_t \leq y_t \forall t \in T$, then the cost allocation of agent i will change to:

$$x_i^{N(\Theta_i)} = \sum_{t \in T} \frac{v_t(\{i\})}{\sum_{j \in N} v_t(\{j\}) - y_t} \cdot (v_t(N) - z_t)$$

Where $N(\Theta_i)$ is the grand coalition with the added knowledge Θ_i .

As the games are sub-additive, it holds that:

$$\sum_{j \in N} v_t(\{j\}) \leq v_t(N)$$

Any information that reduces the costs of a client $v(j) - y$ will reduce the costs of the grand coalition to $\max(0, v(N) - y)$, as costs cannot be lower than 0 in a two price system. The case of $v(N(\Theta_i)) = 0$ and $v(N(\Theta_i)) \neq 0$ can be studied separately. In the case $v(N(\Theta_i)) = 0$, $x_i^{v(N(\Theta_i))} = 0$, and since it is impossible for an allocation to be lower than 0 in the Reduced Cost allocation, $x_i^{v(N(\Theta_i))} \leq x_i^{v(N)}$. In the case $v(N(\Theta_i)) \neq 0$, then it also holds that $x_i^{N(\Theta_i)} \leq x_i^N$. Proof: Given $y_t > 0$, $\sum_{j \in N} v_t(\{j\}) \geq y_t$ and

$$\frac{v_t(N) - y_t}{\sum_{j \in N} v_t(\{j\}) - y_t} < \frac{v_t(N)}{\sum_{j \in N} v_t(\{j\})}$$

Setting $A = v_t(N)$ and $B = \sum_{j \in N} v_t(\{j\})$

$$\frac{A - y_t}{B - y_t} < \frac{A}{B}$$

Eliminate denominators by cross multiplying both sides with $B \cdot (B - y_t)$. Inequality direction is kept because $B \cdot (B - y_t) > 0$ as $B \geq y_t > 0$:

$$(A - y_t)B < A(B - y_t)$$

Expanding both sides:

$$A \cdot B - B \cdot y_t < A \cdot B - A \cdot y_t$$

Subtracting $A \cdot B$ and multiplying by -1 which flips the inequality:

$$B \cdot y_t > A \cdot y_t$$

Dividing both sides by y_t gives:

$$\sum_{j \in N} v_t(\{j\}) \leq v_t(N)$$

Which is true because of sub-additivity.

Because $x_i^{N(\Theta_i)} \leq x_i^N$, then it is always optimal for a player to reveal their true type, meaning the mechanism is DSIC.

An example of how the Reduced Cost allocation looks is in figure 5.4. Agent A and B share the discount created by agent C, while agent C gets their singleton outcome.

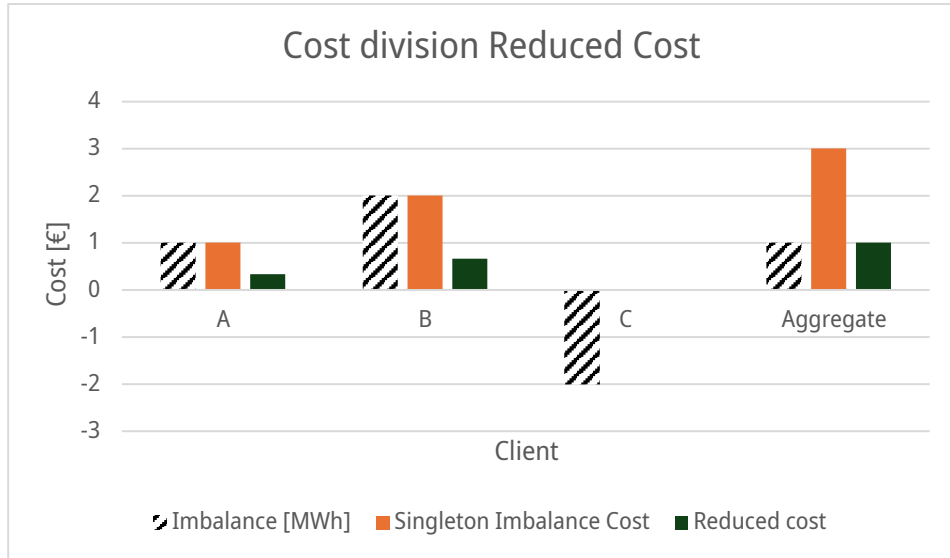


Figure 5.4: Cost allocation for Reduced Cost mechanism

5.4 VCG

Vickrey-Clarke-Groves is another mechanism that will be studied as an alternative to the Shapley Value. VCG is a Grove mechanism, a subset of core-selection mechanisms that are the only ones that are both incentive compatible and efficient [22], therefore it is a useful mechanism where incentive compatibility is needed.

In the VCG mechanism using the clarke-pivot tax, each agent i pays a tax that is the difference between the valuation of every agent except i when part of the grand coalition,

and the total valuation of the coalition without i . This is the externality of i . In the following equation $v_i(S)$ indicates the valuation of i when part of the coalition S . If everyone bids truthfully, then $\sum_{i \in S} v_i(S) = v(S)$.

$$Tax_i(N) = \sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\}) - \sum_{j \in N \setminus \{i\}} v_j(N) \quad (5.6)$$

The payoff that each agent gets is $x_i^N = v_i(N) - Tax_i(N)$. If everyone bids truthfully, then the payoff can be calculated without explicitly calculating the tax, as the marginal contribution of the player:

$$x_i^N = v_i(N) - \left(\sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\}) - \sum_{j \in N \setminus \{i\}} v_j(N) \right) \quad (5.7)$$

$$= v(N) - \sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\}) = M_i \quad (5.8)$$

As this corresponds to the utopian payoff, it is the best payoff that any agent can hope for in the core. The main drawback with the VCG mechanism is that it is not budget balanced, it needs an external party who pays or receives the difference [17].

VCG exhibits additivity, meaning that the VCG taxes will be the same if calculated separately for a set of subgames, or for the main game.

Proof. A sub-game is defined as a game u in a set of games U where

$$\sum_{u \in U} u = v \Rightarrow \sum_{u \in U} u(S) = v(S) \forall S \subseteq N \quad (5.9)$$

The marginal contribution is defined

$$M_i = v(N) - v(N \setminus \{i\}) \quad (5.10)$$

Inserting eq. 5.9 gives

$$M_i = v(N) - v(N \setminus \{i\}) = \sum_{u \in U} u(N) - \sum_{u \in U} u(N \setminus \{i\}) \quad (5.11)$$

Which is the sum of marginal contributions for each subgame. Therefore, VCG is additive.

VCG is much simpler to compute than the Shapley Value, because the tax is only calculated on the basis of the grand coalition N and the coalitions with one agent missing $\{N \setminus \{i\} \mid i \in N\}$, meaning only $|N|+1$ characteristic functions are needed, compared to $2^{|N|}$ for the Shapley Value.

The VCG mechanism is often used in auctions and other non-cooperative games, because there it has a weak budget balance, meaning that the manager never loses money,

resulting from all agents generally having a negative externality in those games [7]. This is not the case in this problem, as there will always be more positive externalities than negative externalities. To properly understand this, it is sometimes helpful to split the VCG payoff into the valuation $v_i(N)$ and the tax $Tax_i(N)$. How exactly the valuation is decided doesn't matter, as long as it is consistent across agents, as valuation and tax will sum up to the same x_i^N no matter the $v_i(N)$, but for the sake of showing budget balance, it becomes easier to see if the valuation is chosen smartly.

Budget balance of cost allocation:

For every realization of imbalance, the sum of taxes paid will be in the range: $0 \geq \sum_{i \in N} Tax_i \geq -\sum_{i \in N} v(\{i\})$, meaning that the manager will never make money.

Proof. Consider the game structure cases:

In the case of *extreme imbalance*, the game is inessential so all externalities will be 0 if clients pay their own cost, therefore no taxes will be paid and VCG gives a budget balanced allocation that is the Uniform Price allocation.

In the case of *high imbalance, dominant direction*, recall from section 4.2.3: $N = \delta^+ \cup \delta^-$ and $\sum_{i \in N} (M_i) = v(N)$. Therefore the allocation where every agent pays their marginal contribution is budget balance.

In the case of *low imbalance*, the game is not budget balanced as the set δ' has a lower marginal contribution than their singleton cost $0 \leq M_i < v(\{i\}) \forall i \in \delta'$. This removes additivity of marginal contributions at the top level. Subsidies increase when net imbalance goes towards 0, peaking at $\sum_{i \in N} Tax_i = -\sum_{i \in N} v(\{i\})$. Take the simple example of two agents with equal and opposite imbalances of 1 and a positive dominant direction. $\Delta_{t,A}^{Real} = 1, \Delta_{t,B}^{Real} = -1$, characteristic functions $v(\{A\}) = 1, v(\{B\}) = 0, v(N) = 0$. If B joins A, they will have $M_B = -1$. A joining B will not change the characteristic function, and A will therefore pay $M_A = 0$. Together, the agents will have a cost of 0, but they must be paid 1 for working together. Each agent can be said to have a positive externality equal to the imbalance cost of the other agent and must therefore be subsidized that amount according to VCG.

In the case of *Non-dominant direction imbalance*, the marginal contribution of all agents are 0, so the allocation will be budget balance.

To summarize, peak subsidies will be found when aggregate imbalance is 0, and will be $\sum_{i \in N} Tax_i = -\sum_{i \in N} v(\{i\})$. Budget balance holds in the high, extreme and non-dominant direction imbalance cases.

Overall, this means that the payoff calculated by the VCG mechanism becomes more budget balanced when there are more agents, agents are comparable in size and imbalance increases, as these factors increase the likelihood of a high imbalance result. For a high imbalance result, the VCG mechanism will return the Uniform Price allocation. For a low imbalance result, the VCG will return a non-budget balanced allocation. For a high imbalance in the non dominant direction, the VCG will return a 0 allocation.

Budget balance of risk assigning:

As VCG is additive, the risk game can be divided into a series of subgames. Assuming that the indices in the tail are always constant:

$$\mathcal{T}_\alpha(J_{t,N}) = \mathcal{T}_\alpha(J_{t,N \setminus \{S\}}) \forall S \subseteq N$$

These subgames can be a game u_t for every $t \in \mathcal{T}_\alpha(J_{t,N})$ where the characteristic function will be $u_t(S) = \frac{1}{|\mathcal{T}_\alpha(J_{t,S})|} \sum_{i \in S} J_{t,S}$, which is the characteristic function of the cost problem multiplied with a constant.

So the risk game can be broken down into a series of cost problems, and will only be budget balanced if all cost problems are budget balanced. Therefore, budget balance does not hold in the general case for the risk problem.

Budget balance can also be broken if an agent is able to change the indices in the α -tail. The details on this are in appendix A.1.

Group rationality for cost problem:

VCG will always return a group rational solution for the cost problem. As VCG is not budget balanced, this allocation is not necessarily in the core.

Proof. As VCG is additive, it will return the same allocation if calculated for every time step separately. As the time steps are independent, the sum of a set of group rational allocations for the sub games will also be group rational for the main game. Therefore group rationality can be checked per time step.

VCG will in the case of *extreme imbalance* and *high imbalance* return the Uniform Price allocation, as for these cases it is the allocation where all agents get their marginal contribution. In section 4.2.3, these have been shown to be group rational. In the *low imbalance* case, the Uniform Price allocation has also been shown to be group rational and will be $x_i = M_i \forall i \in \delta^+ \cup \delta^-$ and $x_i > M_i \forall i \in \delta'$. Since VCG allocation gives $x_i = M_i \forall i \in \delta'$, it will be a cheaper allocation for δ' , even though it is not budget balanced. Because it is the same cost or cheaper for all clients in a group rational allocation, the VCG allocation will also be group rational in this case. For the *non-dominant direction imbalance*, the marginal contribution of all clients will be 0, leading the VCG mechanism to return the allocation of 0 for all clients. As $v(S) \geq 0 \forall S$ in this game, no coalition can have a lower cost than this. As a result of this, the VCG mechanism will always return a group rational allocation for the cost problem.

Group rationality for risk problem:

VCG will also return a group rational allocation for the risk problem.

Proof. The Uniform Price mechanism is found to be group rational for the risk problem in section 5.2. The VCG mechanism will always return a cheaper allocation than the Uniform Price mechanism. This can be seen on a case by case basis: If a coalition has the same α -tail set as the grand coalition so $\mathcal{T}_\alpha(J_{t,S}) = \mathcal{T}_\alpha(J_{t,N})$,

then every index t can be seen as a sub-game, as both VCG and the uniform price mechanism are additive. For every index in the tail that is a *high imbalance* or *extreme imbalance* case, then the Uniform Price and the VCG mechanism will return the same allocation. If any indices have a *low imbalance* case, then VCG will return the uniform price allocation for all agents in δ^+ and δ^- and return the marginal contribution for all clients in δ' , which is cheaper than the uniform price allocation. In the rare case of a non-dominant direction imbalance for a time t in the α -tail, VCG will return an allocation of 0 for all. Because all sub-games are group rational, and VCG is additive, in the case of non-dominant direction imbalance will also be group rational.

If the α -tail sets are not the same, then VCG will provide a discount compared to the Uniform Price. Consider an index t_1 that replaces t_2 in the α -tail of a coalition when client i joins. Client A will have the marginal contribution to that index: $M_{i,t_1} = J_{t_1,N} - J_{t_2,N \setminus \{i\}}$, as t_2 will be replaced by t_1 , instead of $J_{t_1,N} - J_{t_1,N \setminus \{i\}}$, the contribution to costs in t_1 if the index hadn't changed. This gives a discount of $J_{t_1,N \setminus \{i\}} - J_{t_2,N \setminus \{i\}}$ to the client that causes a change in α -tail set.

As the allocations produced by VCG will be the same or cheaper than a group rational allocation for all clients, the VCG mechanism is group rational for the risk problem.

In figure 5.5, an example of cost allocation by the VCG mechanism is given. Each client pays according to their marginal contribution. Because client B and C together have an imbalance of 0, client A adds their singleton cost when joining. Client A and C together have a negative imbalance of 1, so when B joins, their marginal contribution is only 1. This results in a Uniform Price allocation with discounts. Since B only pays 1, the BRP will have to pay for the aggregated imbalance cost of 1€, but receive 0€ from the clients, showing how VCG isn't budget balanced in this case.

5.4.1 Budget balancing VCG

Because of the VCG mechanisms beneficial properties in being incentive compatible and computationally cheap to calculate, there has been interest in remedying its lack of budget balance.

One method proposed by Exizidis[16] is to do an ex-post redistribution of profits based on if a agent contributes to or detracts from budget balance. This is done by solving the market clearing without the agent, and comparing the budget balance with the budget balance when the agent is in the market. If the budget balance of the grand coalition is negative, as it will always be in this game, and the budget balance without the agent is positive, which it will never be in this game, the agent is charged for harming budget balance. Because of this, this fix for budget balance will not work for this problem.

Since there is a limit to how many beneficial properties a mechanism can have, another method is to sacrifice one of VCG's properties in order to gain budget balance. Faltings [17] proposes such a method, where a random agent is designated as a sink. When solving the problem, the sink will then have a lower priority, but will be rewarded by being paid

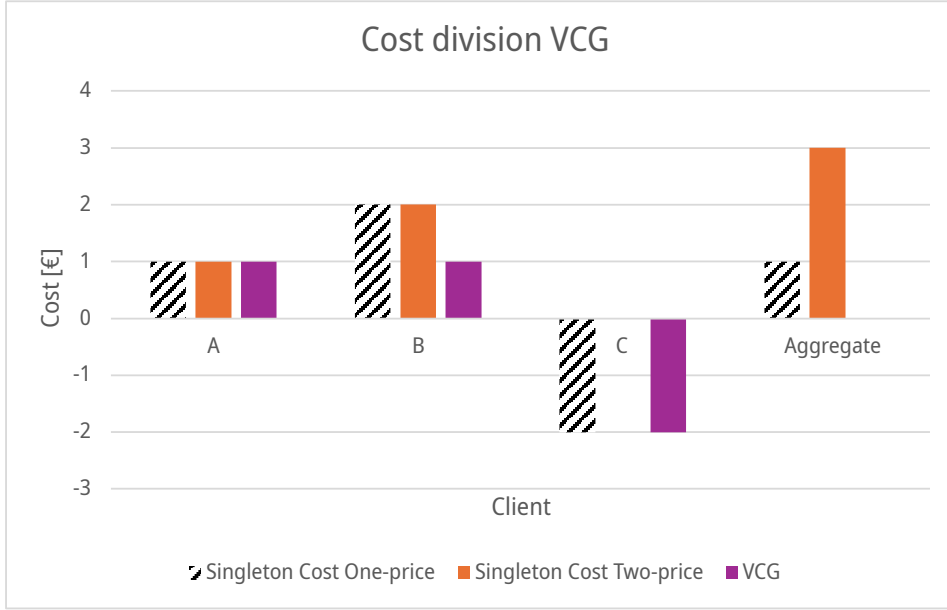


Figure 5.5: Cost allocation for VCG mechanism

the tax the other agents pay. The intuition is that by disregarding the sinks preferences, the other agents can get a better outcome, but they will pay the sink for this. This will result in a solution that isn't efficient, since the sink agent is not accounted for when solving the problem, but this gap can often be smaller than the loss of the agents from having to pay the taxes [17]. This method is not preferable for this problem, as here the agents will always be paid a net subsidy. Bidding according to the sinks preferences can be done without interfering with the bids of the other agents, and since all taxes are subsidies, the sink will have to pay the other agents. Implementing this would reduce the efficiency of the problem, while punishing the sink by making them pay all the subsidies. A method proposed by Parkes [28], is to minimize the distance to the "vickrey discounts", i.e. find an allocation that is as close to the VCG allocation as possible. Here, money is redistributed ex-post to ensure budget balance, while being individually rational and as close as possible to the VCG payment. This method is not directly applicable to this problem, as it relies on a surplus generated by differences between sellers asking prices and buyers bids, but can be adapted slightly to fit this problem:

$$\min_{x_i} \sum_{i \in N} L(x_i, x_i^{\text{vick}}) \quad (5.12)$$

s.t.

$$\sum_{i \in N} x_i = v(N) \quad (5.13)$$

$$x_i \geq x_i^{\text{vick}}, \quad \forall i \in N \quad (5.14)$$

$$x_i \leq v(\{i\}), \quad \forall i \in N \quad (5.15)$$

The objective is to minimize the distance between the budget balanced allocation x_i and the VCG allocation x_i^{vick} via a distance function $L(x_i, x_i^{\text{vick}})$. Constraint 5.13 ensures that the new allocation will be budget balanced. Constraint 5.14 ensures that all clients will pay at least as much in the new allocation as in the VCG allocation. As the VCG allocation is the utopian allocation, any allocation where the client pays less will not be in the core. Constraint 5.15 ensure that the new allocation will be individually rational. The choice of the distance function impacts the distribution. For this problem, the squared relative error will be used: $L(x_i, x_i^{\text{vick}}) = \sum_{i \in N} \left(\frac{x_i^{\text{vick}} - x_i}{x_i^{\text{vick}}} \right)^2$. This is chosen because of the large difference in size of the clients. Taking the absolute difference is likely to advantage large clients, as they will get roughly the same absolute price increase as small clients, it will be a relatively much smaller increase. Note that this is not defined if a client has a $x_i^{\text{vick}} = 0$, so it works best for long time periods.

Whether this changes incentive-compatibility or not will not be proven in this thesis, but in practice, the impact on incentive compatibility is likely to be small, as it would require strategically bidding to be prioritized by the optimization.

5.5 Gately Point

The Gateley point allocates based on the propensity to disrupt of agents, providing a compromise that seeks to keep valuable players in the coalition. Propensity to disrupt is defined as the loss of other players in a coalition caused by agent i leaving divided by the loss to agent i when leaving the coalition.

$$d_i(x) = \frac{\sum_{j \neq i} x_j - v(N \setminus \{i\})}{x_i - v(\{i\})} = \frac{v(N) - v(N \setminus \{i\}) - x_i}{x_i - v(\{i\})} = \frac{M_i - x_i}{x_i - v(\{i\})} \quad (5.16)$$

$$x^N = \arg \min_{x^N \in X^N} \left(\max_{i \in N} d_i(x_i^N) \right) \quad (5.17)$$

Where X^N is the set of all possible allocations in N and d_i is the propensity to disrupt of agent i .

Seemingly, there is an issue when $x_i = v(\{i\})$, as there would be a division by 0. But If the propensity to disrupt for all agents isn't $d=-1$, the the allocation must be $x_i^N = v(\{i\})$ for all agents where $v(\{i\}) = M_i$. [38]

The propensity to disrupt is a measure of the bargaining power of an agent, since it measures how dependent the coalition is of them compared to how dependent the agent is of the coalition. A high $d_i(x)$ indicates that a player has little to gain by joining a coalition, but the coalition gains a lot from them joining. A $d_i(x)$ lower than 1 indicates that the player has more to loose from leaving the coalition, than the coalition has from them leaving. While the thesis problem is a minimization problem, formulation 5.16 can be used for both maximization and minimization problems as the signs will change in both the denominator and the numerator

The Gately Point is the imputation that minimizes the maximum propensity to disrupt, therefore it can be found by solving a nonlinear optimization problem. It has been proven

that this optimal point is the point that makes all players propensity to disrupt equal, and can be found by solving a system of equation or via closed form[38]:

$$d^* = \frac{\sum_{j \in N} M_j - v(N)}{v(N) - \sum_{j \in N} v_j} \quad (5.18)$$

Once d is found, the allocation can easily be found for each client by inserting the optimal propensity to disrupt d^* and solving eq. 5.16 for x :

$$x_i^N = \frac{d^* v(\{i\}) + M_i}{d + 1} \quad (5.19)$$

The Gately Point is more computationally demanding than VCG, as it requires knowing the characteristic function for the grand coalition, the coalitions with one agent missing $\{N \setminus \{i\} \mid i \in N\}$, and the singular clients $\{i\}$. This is a total of $2 \cdot |N| + 1$ compared to $|N| + 1$ for VCG.

Group Rationality:

A standard cooperative game is top dominant if and only if the Gately value lies in the core [20]. A minimization game is standard if $v(\{i\}) \geq M_i \forall i \in N$ and for at least one $j \in N$, $v_j > M_j$. The cost allocation game is standard for all cases except the case of extreme imbalance, as it doesn't fulfill the condition $v_j > M_j$ for at least one $j \in N$ in that case. A minimization game is top dominant if $\frac{v(s)}{|S|} \geq \frac{v(N)}{|N|} \forall S \in 2^N$, meaning that the grand coalition provides the lowest cost per client. This condition is not true in the general case for this problem, for example if a single client causes a very high imbalance cost that isn't countered by others, the average cost of the coalition without him will be lower than the average cost of the grand coalition. Therefore, there is no guarantee that the Gately Point lies in the core of this game or that it is group rational.

A problem with the Gately Point is that it is not well defined for the case where there is an equal propensity to disrupt of -1. In order for the Gately Point to be well defined, Staudacher [38] has found that for weakly superadditive games, one of two conditions must hold. For the equivalent minimization game that is subadditive, the conditions are as follows:

$$v(\{i\}) > M_i \text{ for at least one } i \in \{1, \dots, n\} \quad (5.20)$$

and

$$v(\{i\}) \leq M_i \quad \forall \quad i \in \{1, \dots, n\} \wedge v(\{i\}) > M_i \text{ for at least one } i \in \{1, \dots, n\} \quad (5.21)$$

So either there must be at least one agent whose cost alone is higher than the cost they add to the grand coalition, or if there are no agents that fulfill that condition, at least one agent must add more cost to the grand coalition than their cost alone. The condition $v(\{i\}) > M_i$ is impossible for both the cost and risk problems, so this condition can be ignored. The condition 5.20 will be true in all cases, except the case of extreme imbalance. In the case of extreme imbalance, the allocation $x_i^N = v(\{i\})$ will be the only allocation in the core, as shown in section 4.2.3, so that allocation will be returned instead if the

Gately Point is undefined.

Incentive-Compatibility:

The propensity to disrupt can be calculated as: $d_i(x) = \frac{v(N) - v(N \setminus \{i\}) - x_i}{x_i - v(\{i\})}$. A strategic agent will seek to maximize their propensity to disrupt, in order to be awarded a better allocation in the Gately Point. There are two ways to do this for a strategic agent: Increasing $v(\{i\})$, decreasing $v(N \setminus \{i\})$, or increasing $v(N)$. Since the agent isn't part of $v(N \setminus \{i\})$, they have no influence on it and assuming true costs are known, it is not possible for an agent to increase $v(\{i\})$ by misreporting.

The only path for a strategic agent to increase their allocation is therefore to maximize $v(N)$, which is maximized if every agent tells the truth.

In figure 5.6, the Gately Point allocation is shown. Since the Gately Point is about balancing how much a client can hurt the rest of the coalition by leaving, getting a discount in the Gately Point is dependent on being able to hurt the rest of the clients. Since the coalition B,C are perfectly balanced without client A, client A fulfills the condition $v_i = M_i$, and therefore has to pay their singleton cost because their leaving would not harm the other clients. This causes the Gately Point to mostly give discounts to large clients, because they are the ones who can change the sign of the imbalance by leaving.

In a game where there are many small clients, smaller clients may band together to get the same bargaining power as larger clients. By extending the definition of the Gately Point to consider coalitions as well as individual players, the mechanism is called the *Disruptive Nucleolus* [27]. This is not considered in this thesis.

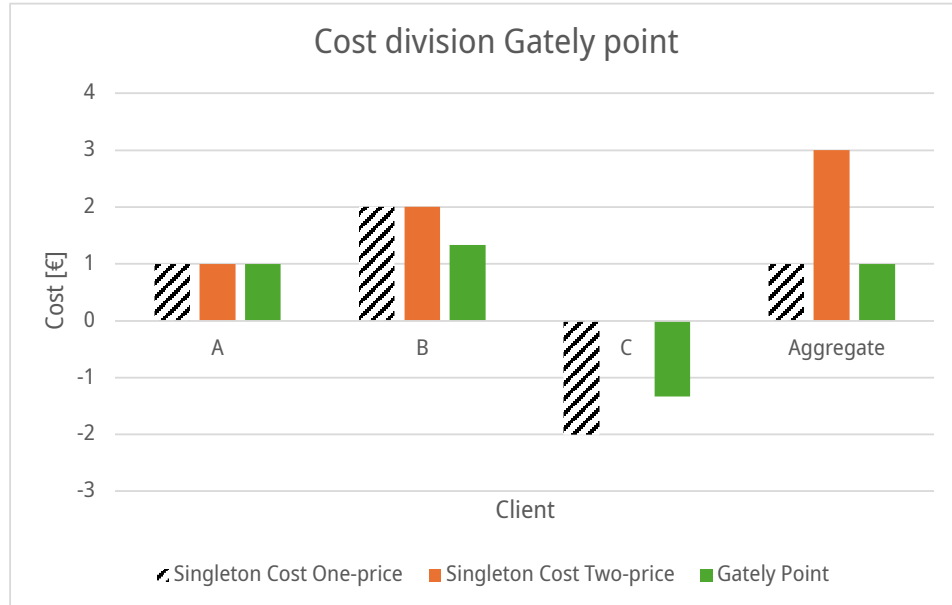


Figure 5.6: Cost allocation for Gately Point mechanism

5.6 Nucleolus

The Nucleolus is a unique imputation that minimizes the excess among agents. The allocation is found by first solving an optimization minimizing the maximum excess, then the excess of the coalition(s) with maximum excess is forced to be constant and the optimization is done for minimizing the second smallest excess. This continues until no more changes are possible. Since the core is the set of imputations where every coalition has a non-positive excess, the Nucleolus will always be in the core if it exists. It will always be in the center of the core, as it is the allocation that is the farthest away from being unstable [27]. In addition, it will be incentive compatible if true costs and utilities are known [40].

To define the Nucleolus, the vector $\theta(x)$ is the excess of every coalition with respect to an allocation x . $\theta(z) \succeq_l \theta(y)$ means $\theta(z)$ is lexicographically larger than $\theta(y)$. This means that sorting from large to small, the first value that is different between the two vectors is larger. E.g. if $\theta(z) = [10, 5, -2]$ and $\theta(y) = [10, 4, 3]$ then $\theta(z) \succeq_l \theta(y)$. The Nucleolus is then the allocation that provides the lexicographically smallest vector of excesses.

$$\begin{aligned} \theta_i(x) &= \epsilon(S_i, x), \text{ where } i = 1, \dots, 2^{|N|} - 1 \\ x &= \{y \in I \mid \theta(z) \succeq_l \theta(y) \forall z \in I\} \end{aligned} \quad (5.22)$$

This low excess is achieved by a series of optimization problems as detailed below. The objective in eq. 5.23 minimizes the maximum excess ϵ by changing the allocation x_i .

Eq. 5.24 sets the maximum excess ϵ to be the highest excess of coalitions that are not in the set Ψ . The set Ψ is the set of coalitions that already had their excesses fixed.

Eq. 5.25 forces the excesses of all coalitions in Ψ to be ψ_S , which is the maximum excess they were assigned in a previous iteration.

Eq. 5.26 ensures budget balance.

After optimizing, the dual value is found for all constraints of the type eq. 5.24. If a dual value for a coalition is not 0, it means that the constraint is binding for that coalition and that the coalition has the maximum excess. All coalitions with binding dual values will be added to Ψ and have the maximum excess ϵ assigned in ψ_S .

$$\min_{\epsilon, x_i} \epsilon \quad (5.23)$$

s.t.

$$\sum_{i \in S} (x_i) - v(S) \leq \epsilon, \quad \forall S \subset N, S \notin \Psi \quad (5.24)$$

$$\sum_{i \in S} (x_i) - v(S) = \psi_S, \quad \forall S \subset N, S \in \Psi \quad (5.25)$$

$$\sum_{i \in N} x_i = v(N) \quad (5.26)$$

$$x_i, \epsilon \in \mathcal{R} \quad (5.27)$$

In Algorithm 1, the algorithm required to find the Nucleolus is detailed. $\mathcal{P}(N)$ is the power set of N , the set of all possible coalitions. The optimization is performed until the

all coalitions are contained in Ψ , requiring $2^{|N|}$ calls of the optimization, if only a single coalition is locked each time.

Algorithm 1 Nucleolus Search

```

1:  $\Psi \leftarrow \emptyset$ 
2:  $\psi \leftarrow \mathbf{0}$  ▷ Initialize excess vector to zero for all coalitions
3:  $x \leftarrow \mathbf{0}$ 
4: while  $\Psi \neq \mathcal{P}(N)$  do
5:    $(\epsilon, S_{\text{locked}}, x) \leftarrow \text{NUCOPTIMIZE}(\Psi, \psi)$  ▷ Find locked coalitions with excess  $\epsilon$ 
6:    $\Psi \leftarrow S_{\text{locked}}$ 
7:    $\psi_{S_{\text{locked}}} \leftarrow \epsilon$ 
8: end while
9: Return  $x$  ▷ Return the allocation  $x$  that results in  $\psi$ 

```

The fact that Nucleolus search is required complicates computing the allocation. Like the Shapley Value, Nucleolus requires knowledge of all $2^{|N|}$ characteristic function. On top of that, the $2^{|N|}$ optimization calls makes the allocation itself complicated to find. Methods like clustered Nucleolus can reduce this complexity by grouping clients together [24], but implementing this is outside the scope of the thesis.

In figure 5.7, an example is given of a Nucleolus allocation. The maximum excess is 0, and is achieved by coalitions $\{A, C\}$, $\{A\}$, $\{B, C\}$. All of these coalitions have the same gain from being a part of the grand coalition, as splitting off. The rest of the coalitions have an excess of -1, indicating that they would rather stay in the grand coalition than split off.

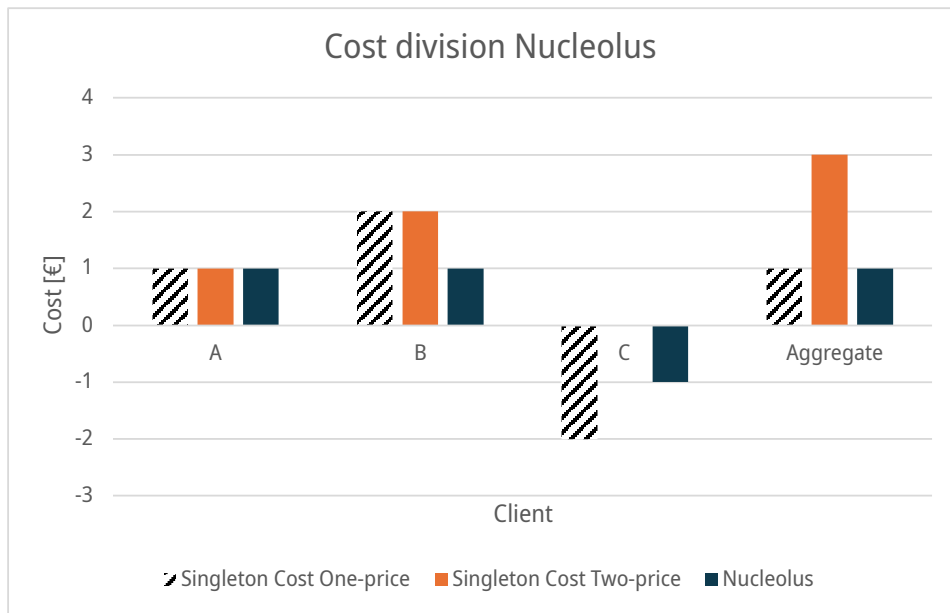


Figure 5.7: Cost allocation for Nucleolus mechanism

CHAPTER 6

Data:

6.1 Data Sources

For this thesis, four datasets of time series data were used, as detailed in table 6.1.

Content	Temporal Resolution	Geographical Resolution	Source
Consumption data	Hourly resolution	Client-level	Reel, Not public
Imbalance price & spot price	15-minute resolution	Price zone level	[13]
PV production	Hourly resolution	Municipality level	[14]
PV forecasts	Hourly resolution	Price zone level	[12]

Table 6.1: Overview of timeseries data sources

Only the price data is natively in 15-minute resolution, all other data sources are changed to 15-minute resolution by assuming an evenly spread production/consumption for the hour. E.g. 4kWh usage for 1:00-2:00 is extended to be 1kWh for 1:00-1:15, and so on.

6.2 Data Availability

The studied period of time is chosen based on the available data. The earliest imbalance price data available in a 15-minute resolution from the TSO Energinet is from 2025-03-04 [13], while the most recent available consumption data for prosumers was 2025-07-19. For the results, a 106 day period from 2025-04-04 to 2025-07-18 is used, to have a month before that can be used for scenarios.

Consumption data availability: Consumption data contains data for the 25 clients that take part in the case study. 22/25 have measurements for every hour, while 3/25 have no measurements in the time period. This missing data is handled by not considering the 3 clients with no data in following analysis.

Imbalance price & spot price: In figure 6.1, the missing data points for price are plotted. Only 4 15-minute periods are missing data values for the imbalance price. This is handled by setting the imbalance price to be the spot price for those periods.

For the two-price scheme, the imbalance direction that is dominating is important, as it dictates in which direction a client must pay the imbalance price, and in which direction they must pay the spot price. 97.56% of the time, these align so an imbalance in the dominant direction means the client loses money, in the remaining 2.44% of cases, the

client will be paid for increasing the imbalance in the dominant direction, according to the dataset. To prevent this dynamic, dominating direction is calculated purely based on the imbalance spread. It is 1 if the imbalance spread is positive, meaning a positive imbalance will lose the BRP money, -1 if the imbalance spread is negative and 0 if there is no need for balancing.

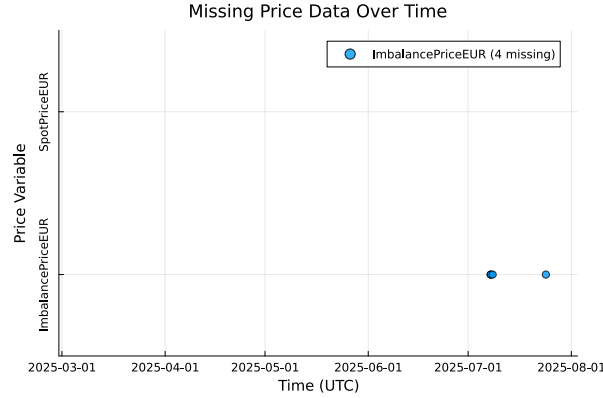


Figure 6.1: Missing data points for price

PV production: production data is on a municipality level, so it is reasonably reflective of the actual production of the PV plant. In order to use it as production data, it is scaled by multiplying it with a factor $k^{\text{PVScaling}} = \frac{P_{\text{case}}^{\text{max}}}{P_{\text{data}}^{\text{max}}}$, where $P_{\text{case}}^{\text{max}}$ is the rated output of the case study PV plant and $P_{\text{data}}^{\text{max}}$ is the maximum production measured at any point in the data. This is under the assumption that the case study PV plant reaches its maximum production at the same time as the other units and that the maximum production of the other units is reached in this period. As the measurement is over the summer period, this is not unlikely.

PV forecast: forecasts are made on an hourly basis on the day before, coinciding with day-ahead market bidding. Scaling is done in the same way as production data, but the forecasts are only on a price zone level, reducing their accuracy regarding the case study plant production.

6.3 Forecasts and scenarios

As forecasting is not a central goal in this thesis, demand scenarios are generated in a simple way. Each scenario is the data shifted by 1 to W weeks. E.g. if calculating for Monday 2 PM with 5 demand scenarios, the last 5 Monday 2 PM demands are used as scenarios.

PV production forecasts are real day ahead solar forecasts from Energinet[12] for the municipality of the PV plant.

Imbalance spread scenarios are 96 data points from midnight to midnight that give the price evolution of the imbalance spread for a full day. U random days are picked for scenarios.

Dominating direction scenarios are generated from the same days as imbalance spread scenarios.

W is set as 5, as it approximates the forecasting model of the BRP in the case study. U is selected as 30, as the in-sample Weighted Mean Absolute Percentage Error (WMAPE), calculated as in eq. 6.1, decreases when increasing U . U is not set above 30 to limit the complexity of the optimization model and reduce the data used for scenarios.

Neither of these methods to set the parameters of the model are ideal. They are used as the focus of the thesis is the game theory, not the modeling. A more appropriate method to find the best values could be a k-fold validation on a validation set, in order to prevent overfitting the models. The in-sample WMAPE depending on choice of W and U can be found in appendix A.2. The highest in-sample WMAPE for W in the range 1 to 9 is 38.3% while the lowest is 36.4%, so the choice of W matters, but the gains of finding the optimal W is not considered a priority, given the focus of the thesis. Increasing U makes a greater difference, but because of limited data availability, will not be set higher than 30. The gains from increasing U are likely to be limited from this point, as the model always bids the predicted net-consumption, as seen in table 6.2 where the error is 0 with perfect forecasts of demand and production.

As a more accurate alternative to scenario based forecasting, a *noise based forecast* is used for comparison. This forecast is not a true forecast, and is not implementable in the real world, as it uses the actual demand data. The forecast is created by taking the actual demand data, and multiplying it with a random normally distributed number with a standard deviation of 0.28 and mean of 1, thereby creating a forecast that is normally distributed around the real value with a standard deviation of 28%. 28% results in a WMAPE of 8.6% of the average demand of the grand coalition, calculated as follows:

$$WMAPE = \frac{\sum_{t \in T} |\Delta_{t,N}^{Real}|}{\sum_{t \in T} P_{t,N}^{D,Real}} \quad (6.1)$$

A WMAPE of 8.6% is reported as a realistic aggregated forecast error for the BRP in the case study. The exact number isn't crucial, as the effects of aggregation will be roughly the same no matter the standard deviation of the normal distributions, as seen in figure 6.2. The y axis is defined $y(\sigma) = \frac{v_\sigma(N)}{\sum_{i \in N} v_\sigma(\{i\})}$, where v_σ is the game resulting from using the noise based demand forecast. The x axis is the relative standard deviation of the demand, a standard deviation of 0.25 meaning that the standard deviation per hour is 25% of the demand for the hour.

The result is that if the forecast is perfect, there are no gains of aggregation, but if the forecast is not perfect, the gains are roughly independent of the standard deviation in the range.

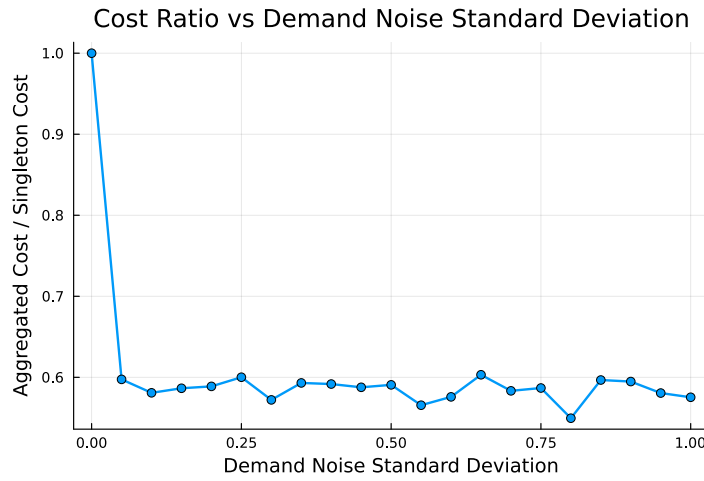


Figure 6.2: Gains of aggregation with noise based demand forecasting and perfect PV forecast, depending on noise standard deviation

6.4 Accuracy of model

The accuracy of the model is measured with WMAPE. In table 6.2 the WMAPE of combinations of demand forecasts is displayed. The measurement is based on the realization of the bids from the optimization model, with scenarios and forecasts given as in the table.

The imbalance spread is also an uncertainty, but in a two-price scheme, the optimization model always bids the expected demand, as evidenced by the perfect demand forecast with perfect PV forecast box having an error of 0%. This could change if the model is made risk-averse, but for this simple model the imbalance spread scenarios are not relevant to the output.

	Uncertain PV Forecast	Perfect PV Forecast
Perfect Demand Forecast	18.1%	0%
Scenarios Demand Forecast	35.1%	26.0%
Noise Demand Forecast	22.9%	8.6%

Table 6.2: Weighted mean absolute percentage error for forecast type combinations

CHAPTER 7

Comparing mechanisms

7.1 Mechanism properties

In table 7.1, the mechanisms are compared across a series of categories. These are sorted roughly in order of importance to this problem. In this section, these mechanisms will be compared based on these categories.

Mechanism	Formula	Computational complexity	Budget balanced	Group rational	Additive	Incentive compatible
Shapley Value	$x_i = \sum_{S \in 2^{N \setminus \{i\}}} \frac{(S)!(N - S -1)!}{N!} (v(S \cup \{i\}) - v(S))$	XXXX	Yes	No	Yes	Yes*
VCG	$v(N) - \sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\})$	X	No	Yes	Yes	Yes
VCG Budget Balanced	Minimize squared distance to VCG allocation	XX	Yes	No	No	No
Uniform Price	$x_i = \sum_{t \in T} \begin{cases} -k_t^{\text{uni}} \cdot \Delta_{t,i}^{\text{Real}} , & \text{if } v(\{i\}) = 0_{**} \\ v_t(\{i\}), & \text{otherwise.} \end{cases}$	X	Yes	Yes***	Yes	No
Reduced Cost	$x_i^N = \sum_{t \in T} \frac{v_t(\{i\})}{\sum_{t \in N} v_t(\{i\})} \cdot v_t(N)$	X	Yes	No	Yes	Yes*
Nucleolus	$\theta_i(x) = \epsilon(S_i, x)$, where $i = 1, \dots, 2^{ N } - 1$ $x = \{y \in I \mid \theta(z) \succeq_I \theta(y) \forall z \in I\}$	XXXXX	Yes	Yes	No	Yes*
Gately Point	$d_i(x) = \frac{\sum_{j \neq i} x_j - v(N \setminus \{i\})}{x_i - v(\{i\})}$, $x = \arg \min_x \left(\max_i d_i(x) \right)$	XX	Yes	No**	No	Yes*

Table 7.1: Studied mechanisms and their formulas and properties

* Assuming true costs/utilities for everyone is known

** Slight difference between cost and risk allocation, see section 5.2.2

*** Group rational only for risk allocation

7.1.1 Computational complexity

In order to use a mechanism, it should be tractable. The group of computationally simplest mechanisms are the basic VCG, Uniform Price and Reduced Cost, as these only require finding characteristic functions for $|N|+1$ coalitions and are very simple to calculate once the characteristic functions are found.

The Gately Point is slightly more complex, as it requires solving $2|N|+1$ coalitions. As is the budget balanced VCG, as it requires solving a simple optimization problem after calculating VCG.

The Shapley Value is much more computationally intensive, as it requires solving $2^{|N|}$ coalitions. This problem has the benefit of being decomposable, so calculating the characteristic function of each coalition can be done quickly by adding together the imbalances of the singleton coalitions and calculating the CVaR. Therefore the Shapley Value remains tractable for large coalitions for this problem. Still, the quadratic scaling causes the Shapley Value to run into memory and computing power issues once coalitions get large, even though calculating the individual coalitions is quite fast. This would be further compounded if decomposeability was compromised, for example by adding batteries.

The nucleolus is the most computationally intensive. It requires solving the same coalitions as the Shapley Value, but finding the nucleolus also requires solving up to $2^{|N|} - 1$ optimization problems in order to minimize the excess. While the scaling is similar to Shapley, the baseline per coalition calculation speed is much lower, leading to hitting scaling issues faster.

7.1.2 Budget balance

All mechanisms except base VCG are budget balanced. VCG is not, and will always require the manager to subsidize the agents. The exact amount of subsidies depend on the game, as detailed in section 5.4.

7.1.3 Group rationality

Group rationality is important for customer satisfaction, and is an indication that the allocation reflects the worth of the agents.

Only the nucleolus has been proven to always be in the group rational for both the cost and risk problem. The Uniform Price allocation will always be group rational for the risk problem, but not necessarily the cost problem. It is possible for a mechanism to return an allocation in the core, even if it isn't group rational, so this doesn't immediately rule out any of the mechanisms.

7.1.4 Additivity

Additivity is relevant in the cost problem, as there will be a subgame for each imbalance realization. Additive methods are preferred, since the choice of period doesn't influence the allocation. Picking non-additive allocations means that time period is a design parameter as well, and the mechanism will behave differently dependent on how often it is recalculated.

7.1.5 Incentive compatible

Only VCG is incentive compatible in the general case, but in this problem the true costs and utilities of agents is known, so only the budget balanced VCG and Uniform Price allocation are not incentive compatible.

7.2 Numerical comparison

Initial numerical comparison is done separately for the cost and risk problem. The numerical comparison is based on the 22 out of 25 clients that have available consumption data. Where possible, all 22 clients are used for results and figures, but some results are too computationally expensive to calculate for the full set, so they are calculated for a subset of clients.

The comparison is made for 106 days from 2025-04-04 00:00:00 UTC to 2025-07-18 00:00:00 UTC. The clients have different demands and demand curves.

7.2.1 Cost allocation

Tested mechanisms

All the mechanisms in table 7.1 are tested. Since the Gately Point is not additive, how many time steps it is calculated over matters. In this section, the "*Gately Point*" will be the Gately Point calculated at once for the entire game, while "*Gately 15Min*" will be the Gately Point calculated for every 15 minute subgame, and then summed together. Nucleolus is not broken up in this way, even though it is not additive, as the computation time would be too long on the available hardware. The budget balanced VCG implementation is not broken up in this way because of time limitations.

In addition, a mechanism called "*Flat Rate*" is included as a benchmark. This mechanism divides the costs so every client pays the same fee per MWh, simulating a case where Reel makes no effort to differentiate between clients. It is calculated in equation 7.1:

$$x_i^N = \frac{v(N)}{\sum_{t \in T} P_{t,N}^{D,Real}} \cdot \sum_{t \in T} P_{t,i}^{D,Real} \quad (7.1)$$

The gains of aggregation

The reduction in costs from aggregation is seen in table 7.2. This difference is the utility gain that must be split among clients when implementing cooperative game theory. This number is likely to change based on the distribution of imbalances from the forecasting method.

A way to understand how a client behaves in the portfolio, is by comparing their singleton costs with grand coalition costs. In figure 7.1, the weighted average ratio of aggregated costs over singleton costs is plotted for portfolios of different sizes, with the y axis calculated as:

$$y(n) = \frac{\sum_{S \subseteq N; |S|=n} v(S)}{\sum_{S \subseteq N; |S|=n} \sum_{i \in S} v(\{i\})} \quad (7.2)$$

where $S \subseteq N; |S|=n$ means all coalitions of length n that can be created from the grand coalition N . The smallest clients, N, W, and X, are left out of figure 7.1 because of hardware limitations, but are included in table 7.2.

The average cost steadily falls as more clients are added. This means that adding more clients doesn't only increase revenue by having a larger customer base, it reduces costs for all existing customers.

Total Aggregated Cost	Total Singleton Cost	Cost reduction
157617€	196756€	19.9%

Table 7.2: Reduction in costs from aggregation, 22 clients

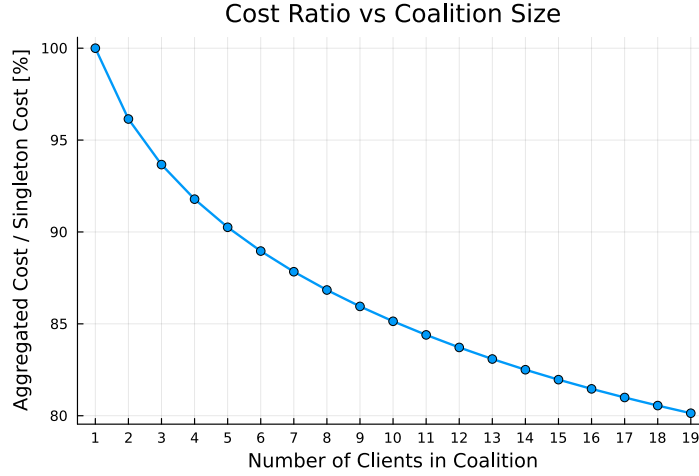


Figure 7.1: Average cost reduction for portfolio sizes, excluding 3 smallest clients

7.2.1.1 Central numerical results and mechanism sorting

As many allocation mechanisms are implemented in this thesis, the most suitable will be found by comparing some initial numerical results, before going into the detailed analysis.

Comparing maximum excess:

In table 7.3, the maximum excess ϵ of the allocations are compared. Excess is calculated based on the full period of 106 days.

Excess is calculated based on two scenarios, one being the 19 largest clients and the other being the 12 largest clients. The 12 client scenario is included as calculating the nucleolus is feasible for this coalition size. Increasing clients beyond 12 makes the nucleolus too slow and memory intensive to calculate on the available hardware. Calculating maximum excess requires calculating all $2^{|N|}$ characteristic functions. Beyond 19 clients, this requires too much memory, so 19 is the maximum number the excess can be checked for in this thesis. The scenarios with 12 and 19 clients respectively do not include the 10 and 3 clients with the lowest demand out of the complete set of 22 clients.

Recall that excess is the difference in utility between the grand coalition and another coalition. Maximum excess indicates how much a client could gain from leaving the grand coalition and joining the best other coalition. All proposed mechanisms provide a negative or very small positive maximum excess, taking into account the full costs of the system, indicating that stability is relatively easy to achieve in this case.

The lowest excess is from the VCG; however, it should be noted that the VCG is not budget balanced, so part of achieving this low excess is subsidizing the clients.

The flat rate allocation is very unstable for this game. Intelligent clients are very likely to drop out of this allocation, and create a coalition with lower costs if they have the possibility.

The nucleolus provides a benchmark for the lowest possible maximum excess. It is lower than any other allocation, but not by a huge amount, considering how computationally

complicated it is. The Gately Point provides the third lowest maximum excess, after the nucleolus and VCG.

Clients	Shapley	VCG	VCG BB	Gately Point	Gately 15 min	Uniform Price	Reduced Cost	Flat Rate	Nucleolus
19	4.7€	-132.2€	-0.1€	-36.2€	-16.0€	-16.2€	2.6€	15105€	–
12	-63.0€	-789.7€	-0.6€	-220.9€	-188.0€	-81.9€	98.0€	10868€	-394.8€

Table 7.3: Maximum excess of allocations for cost

Computational complexity

All computations are done on a AMD Ryzen 5 7535HS CPU with 30,8 GB of usable RAM and a 54 GB paging file on a SSD. The computations are not parallelized.

In figure 7.2 and table 7.4, allocations are compared for up to 12 clients. In figure 7.3, nucleolus is left out, allowing calculations to be made for the rest of the clients. Timing results for 22 clients are in the appendix A.3.

Nucleolus is by far the most computationally intensive method. The computation time for the Shapley Value scales approximately at the same rate as the nucleolus, but the nucleolus has a much higher base computation speed.

The source of computational complexity are two-fold: Calculating characteristic functions and calculating allocation.

Calculating the characteristic function is the main source of complexity for most of the mechanisms. The mechanisms can be divided into two groups, those of low complexity and those of high complexity. High complexity mechanisms require characteristic functions for all $2^{|N|}$ coalitions, while low complexity mechanisms require less.

Even though the Uniform Price, Reduced Cost and VCG mechanisms only needs $|N|+1$ coalitions, it is computationally faster to optimize N times and add them up, than to optimize $|N|+1$ times. The Gately Point requires $2|N|+1$ characteristic functions, which can also be found by optimizing $|N|$ times and adding them up. The time to sum them is negligible, therefore the characteristic function computation time is the same as for all the low complexity mechanisms.

Scaling up to 22 clients increases the characteristic function calculation time to 1.1 seconds for the simple mechanism and 71 seconds for the complex, indicating how the simple mechanisms scale much better. Shapley ends up taking 152 seconds to calculate for a day with 22 clients, with 71 seconds being characteristic function calculation and 81 seconds being allocation calculation. Both the characteristic function and allocation calculation could be sped up, as the current implementations are slow and simple for-loops because the limited available RAM made faster methods impractical. While Shapley is feasible for 22 clients and could be optimized to be feasible for larger sets, its quadratic scaling will make it intractable much earlier than the simple mechanism.

The nucleolus differs from the other mechanism by its allocation calculation being much slower than characteristic function calculation. This is a result of having to optimize excess up to $2^{|N|}$ times.

Budget balance:

In table 7.5, the budget balance of VCG is shown for the full 22 client game. The

Allocation	Characteristic function time [s]	Allocation time [s]	Total time [s]
Shapley	0.33	8.6×10^{-3}	0.34
VCG	0.29	4.4×10^{-6}	0.29
VCG Budget Balanced	0.29	1.2×10^{-3}	0.29
Gately Point	0.29	6.4×10^{-6}	0.29
Gately 15 Min	0.29	2.9×10^{-3}	0.29
Full Cost	0.29	4.9×10^{-4}	0.29
Reduced Cost	0.29	3.9×10^{-4}	0.29
Nucleolus	0.33	85.0	85.3

Table 7.4: Computation Times for Coalitions with 12 Clients (Scientific Notation, 2 Significant Digits)

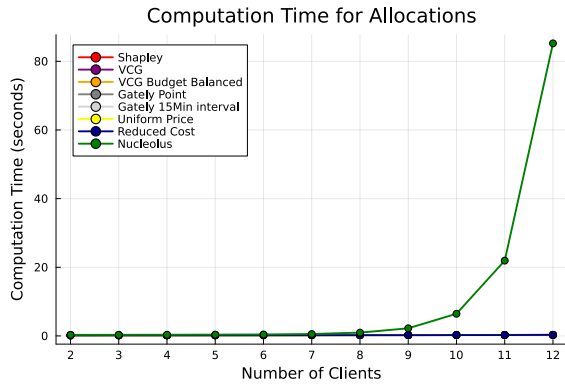


Figure 7.2: Computation time of allocations for a day

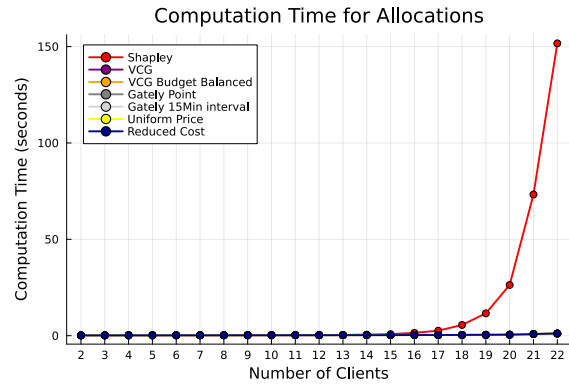


Figure 7.3: Computation time of allocations excluding nucleolus for a day

subsidies given are significant, and are likely to increase if forecast accuracy increases, as found in section 5.4. All other mechanisms are budget balanced, so they are not shown here.

Total Cost	Total VCG Cost	Unassigned Cost
157617€	142919€	9.3%

Table 7.5: Budget balance of VCG for cost problem, 22 clients

Picking the best candidates for cost allocation

To improve the legibility of the figures in the in-depth study of how the allocations affect individual clients, a few will be chosen as the prime candidates.

The Shapley Value is feasible for calculating a day at a time for 22 clients, but because of hardware limitations it is too time consuming to calculate for the whole 106 day period, and therefore it is excluded from further analysis.

VCG has a significant budget imbalance for this application, and since so many other mechanisms are incentive compatible for this game, VCG is not seen as viable.

Nucleolus is excluded from further analysis in this thesis, as it is seen as infeasible to

calculate for the full 22 clients.

That leaves the Gately Point calculated for the full period and every 15 minutes, the Uniform Price, the Reduced Cost, the budget balanced VCG and the flat rate as a benchmark.

7.2.2 Risk allocation

This numerical comparison uses the same input data and models as section 7.2.1. The main difference is that instead of having the cost as the characteristic function, how each client contributes to CVaR is allocated instead. The α parameter is set to 0.05 for this section.

The allocation of CVaR is different from cost, as the BRP can not directly tell a client that they must pay this amount. For this metric to be useful, the BRP must have some equation that incorporates their aversion to risk and the α in order to return the price of 1€ of CVaR.

Tested mechanisms:

Fewer allocation methods are implemented for the risk problem compared to the cost problem. Shapley, VCG, budget balanced VCG, the Gately Point, Uniform Price, and the nucleolus are applied to the problem. The Gately Point is only computed for the whole period, with no alternative 15 minute interval computation. This is done because the CVaR metric by design needs to cover multiple time periods. A risk centric Reduced Cost allocation is not implemented because of time constraints.

The gains of aggregations:

To show the gains of aggregation for risk, the aggregated CVaR is compared to the sum of individual CVaR's. This is an abstract metric, as CVaR cannot usually be summed in this way. This metric is used since the alternative to applying cooperative game theory to risk is to measure the CVaR of each client separately, and assign costs based on that. In table 7.6, the difference between assigning CVaR based on cooperative game theory and based on singleton costs is 11.3%.

Aggregated CVaR	Sum of CVaR	Difference
180.3€	159.9€	11.3%

Table 7.6: Difference between individual and aggregated CVaR, 22 clients

7.2.2.1 Central numerical results and mechanism sorting

Like for the cost problem, not all allocations are suited for the risk problem. Here, all of them will be compared and the prime candidates will be picked.

Comparing maximum excess:

In table 7.7, the maximum excess for allocations applied to the risk problem is compared.

Note that the values are much smaller than for cost, as this is the change in CVaR. These are also not calculated for the full 22 clients, because of hardware limitations.

All allocations except the flat rate provide a stable or almost stable allocation. The Gately Point is the most stable among the allocations except the nucleolus and VCG, but the differences are quite small.

Clients	Shapley	VCG	VCG BB	Gately Point	Uniform Price	Flat Rate	Nucleolus
19	0.02€	-0.21€	0.001€	-0.058€	-0.0005€	14.9€	–
12	-0.08€	-1.04€	-0.0005€	-0.29€	-0.0014€	10.72€	-0.52€

Table 7.7: Maximum excess of allocations for CVaR

Computational complexity:

The computation speed is very close to the same as for the cost problem, as the allocations are the same, except characteristic function calculation speed is slightly longer because of the need for calculating the CVaR of the coalitions. The results for 12 and 22 clients can be seen in appendix A.4 and are similar to the cost problem, with the Nucleolus being too computationally expensive for the full client set, and the Shapley Value being too computationally expensive to calculate for the 106 day dataset.

Budget balance:

In table 7.8, the budget balance of the VCG mechanism when applied to CVaR for the dataset of all 22 clients is shown. About 20€ of CVaR risk is unaccounted for when applying VCG. That risk must not be taken into account when allocating risk based on VCG, in order to preserve its properties.

Total CVaR	VCG CVaR Total	Unassigned CVaR
181.1€	160.6€	11.3%

Table 7.8: Budget balance of VCG

Picking the best candidates:

Because of their long computation times, the Shapley Value and the Nucleolus are seen as impractical for this problem as well. Likewise, VCG is also seen as impractical because of its lack of budget balance and that budget balanced incentive compatible alternatives exist.

That leaves the budget balanced VCG, the Gately Point, Uniform Price and the flat rate benchmark for further study.

7.2.3 In Depth Analysis

In this section, the best candidates for mechanisms will be compared across the portfolio for both cost and risk assigning, to understand how they behave with respect to individual clients.

Demand of clients

In figure 7.4, the total demand for each client in the period is shown. In this case, there is a very large difference between the biggest and smallest clients, with client A having a demand that is larger than all other clients combined. For subsequent plots, clients are sorted in this order, according to their total demand.

In figure 7.5, the PV Coverage defined as the total PV production divided by demand for each client is plotted, $PVCover = \frac{\sum_{t \in T} P_{t,i}^{G,Real}}{-\sum_{t \in T} P_{t,i}^{D,Real}}$. A few small clients have very high PV production compared to their demand in this period. This plot shows how much uncertainty comes from the demand vs PV production for a client. Client K's cost and risk will mainly come from forecasting demand for purchasing electricity, while client F's cost and risk will mainly come from forecasting PV production to sell electricity. As the portfolio effect comes from forecasting errors cancelling each other out, and the PV forecast being the same for each client, clients with a high PV coverage will likely benefit less from cooperation.

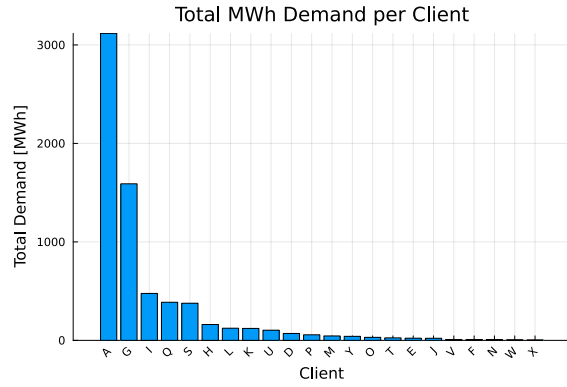


Figure 7.4: Demand of clients in the 106 day period

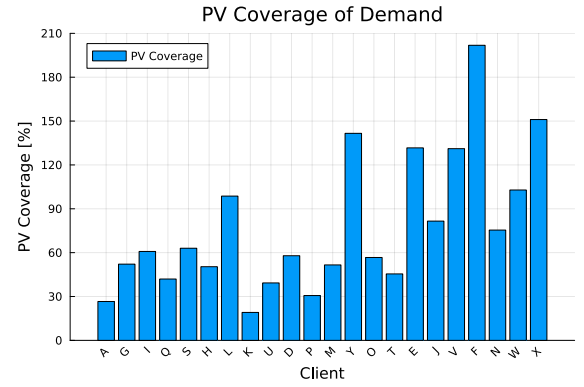


Figure 7.5: Total PV production as percent of total demand

Total allocation to Clients

In figure 7.6 and 7.7 the total cost and CVaR allocated to each client is compared. From comparing figure 7.4 and 7.6, it is clear that the imbalance costs of clients has a strong relationship with the clients demand. What is interesting from these figures is how the allocation mechanisms cluster together. Every allocation mechanism except for the flat rate allocation is clustered close together, indicating that moving from flat rate to any other allocation, provides a large improvement in fairness, but differences between allocations are minor in comparison.

Client A and G seemingly overpay quite a bit with the flat rate allocation, while client I underpays.

In figure 7.7, the total CVaR allocated to each client is plotted. α is set to 0.05, so the CVaR is the average cost in the worst 5% of 15 minute imbalance settlement periods. The shape is similar to the plot for cost in figure 7.6, with a CVaR contribution that is closely tied to demand. Allocations are slightly different, e.g. for client A.

This figures gives an overall view of who the flat rate allocation benefits and who it harms,

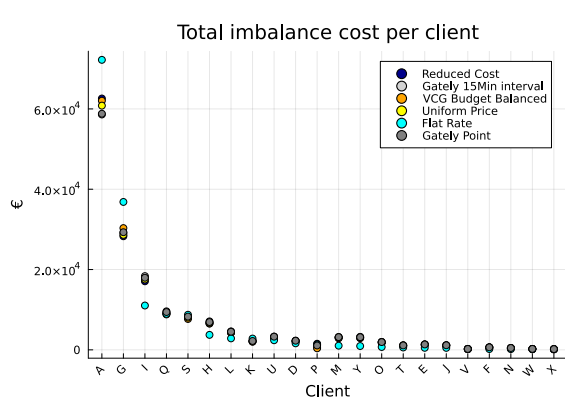


Figure 7.6: Cost allocations for each client

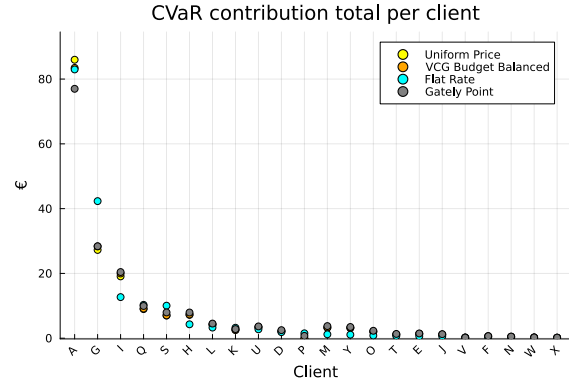


Figure 7.7: CVaR allocations for each client

but to see the differences between allocations and the smaller clients, other visualizations are needed.

Cost Per MWh

As the differences in allocations among clients are difficult to see except for the largest clients in figure 7.6 and 7.7, figure 7.8 compares the imbalance costs per MWh demand. The y axis is defined for both cost and risk as: $y_i = \frac{x_i^N}{\sum_{t \in T} P_{t,i}^{D,Real}}$.

This makes it more clear who are the clients that cause the highest imbalances. Even though clients A and G have high total costs, as seen in figure 7.6, their cost per MWh is lower than most other clients.

Client P is another client with a relatively low cost, but there is a large difference in how the individual allocations value client P. The Gately Point calculated on a 15 minute basis and the budget balanced VCG attributes a very low cost to the client, while Reduced Cost attributes a higher cost.

The clients that have lower demands than P generally have higher imbalance costs. This could be influenced by them having a PV coverage. The difference between clients is large, highlighting the need for individualized fees. Client Y has a roughly 4 times higher contribution to cost and risk than P, depending on chosen allocation.

In figure 7.9, the CVaR contribution is divided by total demand in the period in the same way as cost. This figure gives an indication on how much each client should be charged on top of the spot price, to account for the risk they add to the BRP.

Figures 7.8 and 7.9 have similar trends with slight differences. There is a correlation between a client having a high cost and risk in a two-price system. It makes sense that more expensive clients are generally more risky, especially considering that both cost and risk relies on aggregation reducing the variance of the portfolio. The most significant difference is that a client can have a negative contribution to risk, as seen for client P.

Individual Client Cost Differences

When relative to demand size, the allocations are still clustered quite close together compared to putting a flat rate across all clients. While going from a flat rate to some

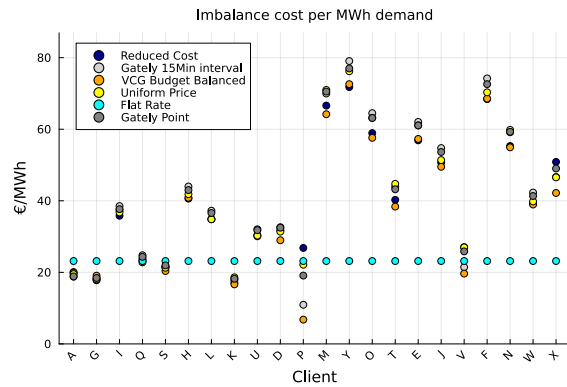


Figure 7.8: Imbalance cost per MWh of demand in 106 day period

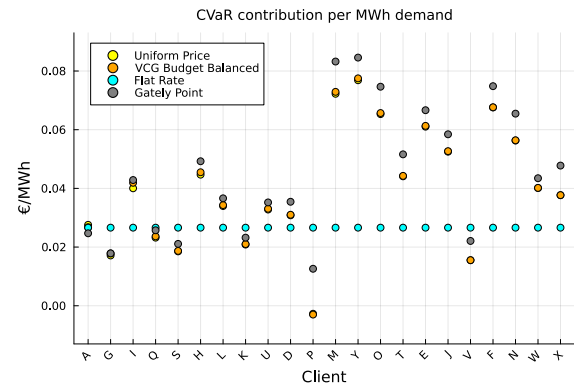


Figure 7.9: CVaR contribution per MWh demand

individualized allocation makes the largest difference, there are also significant differences between allocations. In figures 7.10 and 7.11, the cost increase for each client going from their cheapest to most expensive allocation is shown. While the differences in maximum excess from table 7.3 are small, there is a large difference between who gets a good deal between allocations. E.g. client P would greatly prefer a Gately Point solution calculated on a 15 minute basis or the budget balanced VCG, compared to Reduced Cost, as Reduced Cost is more than three times as expensive. The preferred allocation for each client can be found in appendix A.5. Budget balanced VCG and Gately 15Min are often at the extremes of being the most expensive or cheapest for clients. The Gately Point calculated for the full period is notable for only being the most expensive or cheapest allocation for client S, making it more of a middle ground compared to other mechanisms. This plot is only made for the cost problem, but the risk problem has similar results, as seen in figure 7.9.

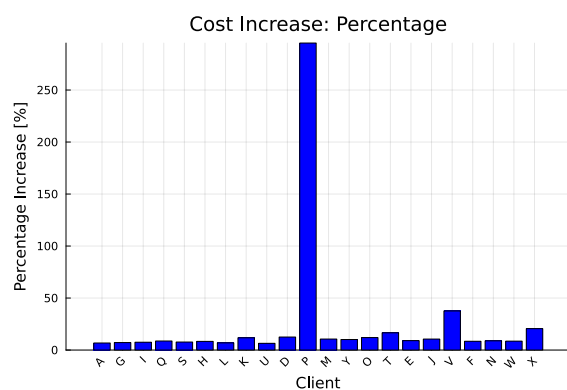


Figure 7.10: Percentage increase from cheapest to most expensive allocation

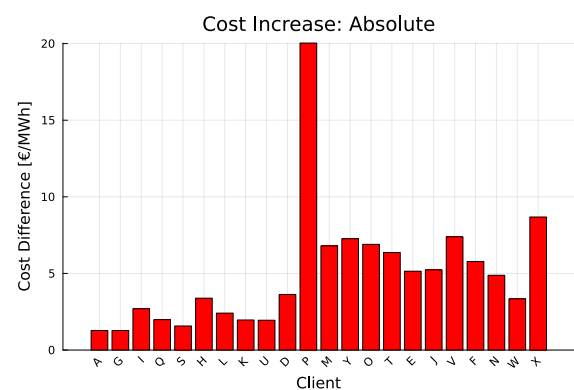


Figure 7.11: Absolute increase from cheapest to most expensive allocation

Cost ratio of clients

A different way to understand how a client behaves in the portfolio, is by comparing their singleton cost with grand coalition allocation. This will be called their relative cost

or relative CVaR, defined in this thesis as:

$$z_i = \frac{x_i^N}{v(\{i\})} \quad (7.3)$$

This is a measurement that indicates how well a client works as part of the portfolio. A relative cost of 100% means that the client has the same cost as if they were not part of the portfolio, while a low relative cost means that they synergize well with the rest of the portfolio. A client with a low relative cost is more likely to be satisfied with being a part of the portfolio, as their costs are lower than they would be alone.

This is illustrated for cost in figure 7.12. Interestingly, most clients in the portfolio have a relative cost of around 80%, even though clients have large differences in costs in figure 7.8. Clients P and V have notably lower relative cost than the others, indicating a different pattern in demand that works better in the portfolio than the demands of other clients.

The flat rate relative cost varies greatly. This is not indicative of the portfolio effect on the clients, as the flat-rate doesn't reflect the true worth of the clients, evidenced by its high excess.

The results are again similar but slightly different for risk. In this case, the budget balanced VCG and Uniform Price give lower costs to smaller clients. Interestingly, the budget balanced VCG and Uniform Price return almost the same relative cost for CVaR, while there is a larger difference between them for the cost problem.

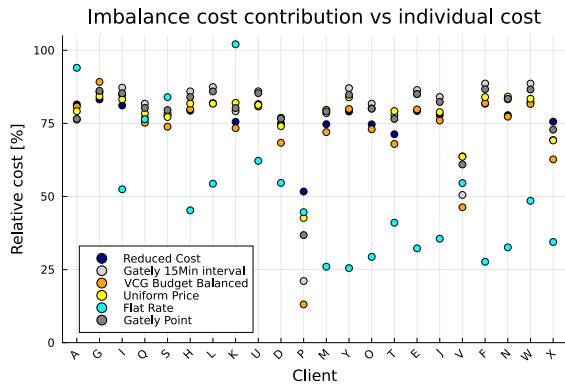


Figure 7.12: Costs as part of coalition vs alone

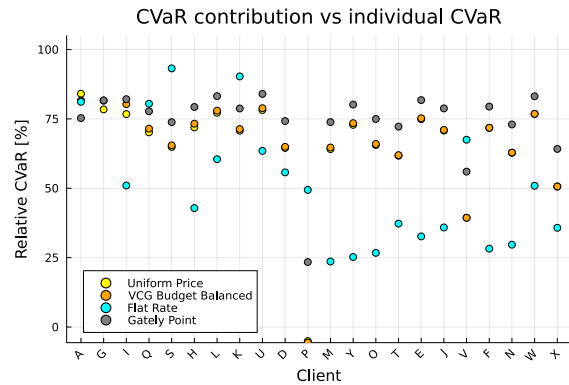


Figure 7.13: CVaR as part of coalition vs alone

Daily variability in results for cost

The previous results have been made with the assumption that an entire 106 day period is assessed at once. A different model could be that costs are reflected daily. This makes less sense to do for risk, as determining risk ex-post is sensitive to time period.

In figure 7.14, the allocated daily costs are compared to the daily singleton costs for client A and P using the relative cost metric. A is chosen as an example of a large client and P is chosen as an example of a client with a large difference in allocation preference. The black line is the median cost per day, while the blue line is the mean cost weighted

by daily imbalance, which corresponds to the dots on figure 7.12. For this plot, budget balanced VCG and the Gately Point are calculated based on a day. Because they are not additive, the results are slightly different from those in 7.14. All else being equal, a mechanism that provides a low variability is preferable, as it is less risky for the clients. Clearly, client P experiences a much wider spread of relative costs, if outliers are not taken into account. The variances of the allocations are similar, however, Reduced Cost mechanism stands out among the others as having an especially low variability. This is to be expected, as by design no client will ever be paid for participating in this mechanism, so relative costs will be locked between 0% and 100%.

No client ever pays more than their singleton costs for any of the allocations, satisfying individual rationality.

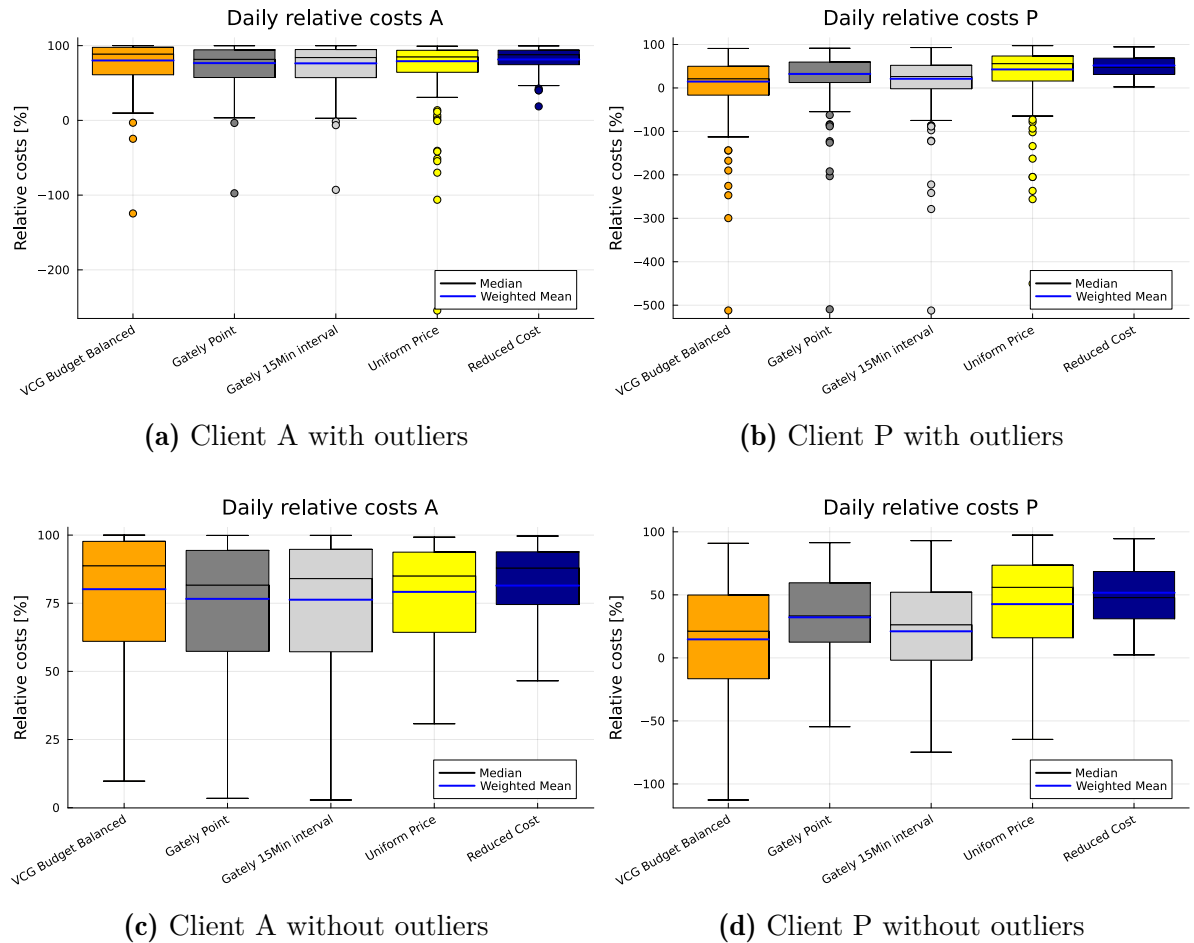


Figure 7.14: Boxplots of daily costs for clients A and P, with and without outliers

7.2.4 Effect of forecast accuracy on cost results

In this thesis, the focus has been dividing the costs and risk, not generating the bids. As such, there are significant errors in the demand and PV forecasts as noted in section 6.4. To give a perspective on how the forecasts affect the allocation, a forecast based on normally distributed noise as explained in section 6.3 and an assumption of a perfect PV forecast are presented. This reduces the WMAPE of the grand coalition from 35% to 8.6%, and removes any differences in predictability of clients. Figure 7.15 displays the effect this has on cost and table 7.9 compares the maximum excess of mechanisms in the cost problem under this assumption.

By removing PV related uncertainty and making the uncertainty of demand normally distributed, the only differences between clients is the timing and magnitude of their demand.

In figure 7.8 there was no direct connection between client size and allocation, but when removing other sources of uncertainty, there is a clear trend. Client A, who is larger than the rest of the clients combined, pays a much higher fee than the rest. The intuition behind this is: an imbalance in the dominant direction is less costly if other clients can counter it by an opposite imbalance, and an imbalance in the non-dominant direction is worth more if it can counter an opposite imbalance.

If a large client like A has an imbalance, it is likely larger than the rest of the imbalances combined, so their imbalances will only partially be countered or partially counter the imbalances of the rest of the clients.

Although the budget balanced VCG is stable for scenario forecasting, it provides a large positive excess in table 7.9. This is also apparent from figure 7.15b, where client A will pay their singleton costs under this allocation, giving them a high chance of getting a better allocation somewhere else. For this reason, this implementation of budget balanced VCG is not considered a good mechanism as forecasts get better.

Another interesting result from this, is the spread between the allocations. From table 7.9, the allocations except budget balanced VCG provide a stable or very close to stable allocation, but figure 7.15a reveals that there is a large difference between who it benefits. The budget balanced VCG and Gately Point calculated in 15 minute intervals will prioritize the smaller clients, consistently giving them the lowest costs while giving client A a high cost. The simple methods, Reduced Cost and Uniform Price are the opposite, giving higher costs to small clients and lower costs to large clients.

The takeaway from this is that the accuracy of the forecast and how the forecast errors are distributed have a great effect on the allocations. As overall imbalance falls, the choice of allocation becomes important. Client A is unlikely to be happy with the 15 minute Gately Point allocation, even though it is fair according to cooperative game theory.

Forecast type	VCG BB	Gately Point	Gately 15 min	Uniform Price	Reduced Cost	Flat Rate
Scenario forecast	-0.1	-36.1€	-16.1€	-16.2€	2.6€	15104€
Noise forecast	144.8€	-21.8€	-11.3€	-22.2€	-21.2€	5626.8€

Table 7.9: Maximum excess of allocations comparing forecast types, 19 clients

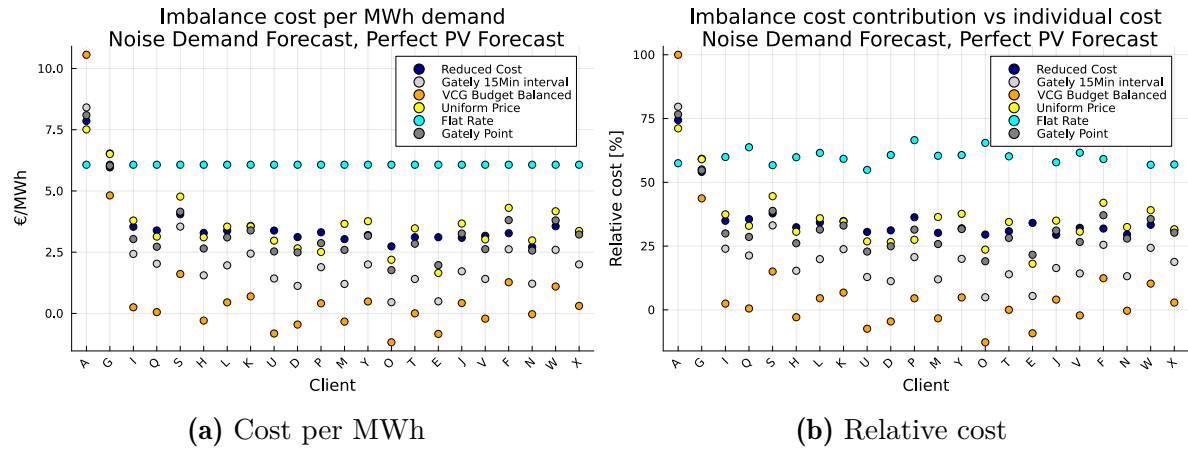


Figure 7.15: Cost metrics for model with perfect PV forecast and normally distributed error in demand forecast

7.3 Summary of results

Based on their properties, computational complexity and maximum excess, the most suitable allocations for allocating costs among those studied has been found to be: Reduced Cost, Uniform Price and the Gately Point. For assigning risk, the most suitable allocations have been found to be: Uniform Price and the Gately Point.

The Gately Point provides the lowest maximum excess in many cases, and is therefore the most fair of the allocations; however, the maximum excess of the other allocations is also acceptable and not far from the Gately Point, so the choice between them could come from other factors.

An important factor for a BRP could be transparency. In cooperative game theory, it is assumed that clients are intelligent, i.e. they understand the allocations and can calculate everyone's utility and excess. In practice, the Reduced Cost and Uniform Price allocations are much easier to understand for a client, so they may be better tools to negotiate fees with. This also makes them easier to maintain, as they don't require specialized knowledge.

Another factor is who the allocation prioritizes. As shown in figure 7.11, the difference between stable allocations can be quite large, so the choice of allocation is also a choice of which clients to give a pricing advantage. A BRP might seek to prioritize large clients over small clients, as they require similar work to forecast but have much higher volumes. Based on figures 7.8 and 7.9, the Gately Point generally gives better allocations to the large clients than the small clients, so it could be a preferable choice if wanting to give more competitive deals to them. The noise based demand forecast in figure 7.15a results in the Gately Point prioritizing smaller clients, so the exact clients allocations favor rely on more complex dynamics than pure demand size.

CHAPTER 8

Discussion

Throughout this thesis, various allocation mechanisms have been studied and compared in relation to the problem of allocation imbalance costs for a BRP in a two-price market. In this section, the weak points and potential paths forward for this thesis are discussed.

8.1 Points of Critique

8.1.1 Forecasting and Model Accuracy

This thesis has provided various methods to divide costs, but has not sought to study how to reduce it. The simple scenario based forecasting coupled with the optimization model provides a relative MAE of 35.1% throughout the studied time period. The forecast has a large effect on the allocation, as seen in section 7.2.4. A more accurate forecast leads to larger differences between the allocations, so the choice becomes more significant as forecasts get better.

This thesis provides some general tools for how to allocate costs in a portfolio, but this case study cannot be used directly to allocate costs to the studied portfolio, as the costs and differences between clients depend greatly on the forecasting methods used by the BRP.

A related issue is that this thesis only studies the day-ahead market. In practice, a BRP will trade in multiple markets in order to reduce their costs. Only looking at the day-ahead market further exacerbates the problem of inaccurate forecasts, as it exposes the BRP to the full cost of their inaccuracy with no way of remedying it in later markets. Trading in more markets is likely to complicate the generation of the characteristic functions, as the trading actions and costs depend on the imbalance of the coalition and how forecasts evolve, leading to having to calculate characteristic functions for coalitions independently, instead of summing singleton costs.

8.1.2 Implementation

Another point is on whether these techniques are practical to implement. They require a fundamental shift in the pricing model and work done by the BRP, so it could be put into question if the implementation is worth it. There are two primary complications relevant to this:

Firstly, calculating the allocations requires knowledge of the imbalances caused by every single coalition. This requires forecasting the demand of every client individually, which adds complexity to designing the forecasting models compared to having a single aggregate forecast. If only considering the day ahead market and using a simple model where the bids equal the forecasts, then the BRP can calculate how the imbalance would resolve for each coalition, by adding together singleton imbalances, as in this thesis. This becomes complicated if more markets than the day ahead market are considered. In that case, the BRP must have some method to simulate what the cost of the coalition would be if trading in the individual markets. Trading individually for each client, treating them as if there were no other clients in the coalition and summing up costs to create coalitions is likely to give a bad approximation. If there is a spread between the price of buy and sell orders on the intraday market, adding together two clients with equal but opposite imbalances would have a net cost, where their aggregated costs would be 0. A better method would be to create a simulation model under price-taker assumptions, that simulates what the costs would have been for every coalition. This adds another layer of complexity to the implementation.

Secondly, it requires work to implement in the BRP's business model. The simplest method would be to calculate it ex-ante, working it into the flat fee on top of spot price, that is offered to the clients upon signing the contract. This requires that prospective clients give access to historical demand data before signing the agreement. Another model is to introduce a new product that reflects the cost of the clients imbalance ex-post, but that requires competitive pricing to entice the client to take on the risk.

8.1.3 Risk Metric

CVaR is used as a risk metric in this thesis. CVaR can be problematic if calculated for short periods ex-post, as an otherwise very predictable client could be unfortunate and have a dominant direction imbalance in an expensive period, while unpredictable clients could be lucky to only have high imbalances in cheap periods. This accurately reflects the risk the client added in the measured period, but doesn't necessarily predict the risk the client adds in future periods. An alternative could be to base the characteristic function for risk on imbalance volumes. This can be a better predictor of future risk, but a worse measurement of the cost of the client ex-post.

8.2 Further Work

8.2.1 Repeated Game Stability

In cooperative game theory, it is assumed that clients will accept an allocation as fair if they cannot get a better payoff by making another coalition. This can lead to allocations where some clients have little reason to cooperate, like the uniform price allocation where a set of clients will pay their singleton costs, the worst possible allocation for them.

This is a very static view of stability, as in the real world clients have more nuanced

options than creating a smaller coalition or leaving to work alone.

Both for the cost and risk problem, there is a large range between the best and the worst allocation for each client. A client might seek to negotiate, stating that they will leave the cooperation unless a different allocation mechanism is used. This might be an immediate loss for the client, and therefore it will not happen in classical game theory, but the client might seek new coalitions elsewhere.

8.2.2 Individual Client Studies

From section 7.2.1, there are some clearly good and bad clients. Client P and V were found to be especially cheap to have as part of the portfolio. It could be useful to develop a heuristic for the worth of clients by studying what makes a client "good" and what their cost and risk will be, to be able to circumvent the complete game theory and provide a more accurate price ex-ante.

8.2.3 Including Batteries

A connected use case is the sharing of batteries. In this thesis, clients share ownership of a PV plant. The only complication this causes is that the imbalances caused by the PV plant will be in the same direction for all clients. A place where cooperational game theory could be especially important is if clients invest in a battery together. In this case, the batteries will be best used as a shared resource among clients, necessitating optimizing all characteristic functions individually. This will make the complex mechanisms infeasible, but will leave the simple mechanisms as alternatives.

CHAPTER 9

Conclusion

This thesis set out to answer the question: **How can the true imbalance costs of prosumers in a portfolio be allocated using cooperative game theory?**

The analysis confirms that aggregation reduces both cost and risk for Balance Responsible Parties, but introduces the challenge of fair redistribution among diverse clients.

By applying cooperative game theory to a real-world portfolio, this work demonstrates that, while classical solutions like the Nucleolus and VCG offer theoretical elegance, their computational and budgetary limitations hinder real-world application. The Gately Point and simpler mechanisms such as Reduced Cost and Uniform Price emerge as viable alternatives, balancing tractability and stability.

Stability was found to be possible to achieve for most mechanisms, but stability does not ensure fairness. Even among stable allocation, there are large differences in costs allocated to clients, with the most expensive allocation being more than twice as expensive as the cheapest in some cases. This issue is magnified as forecasts improve, widening the gap between allocation, making the mechanisms bias towards certain client types an important consideration when choosing mechanisms.

Importantly, the findings emphasize the role of client diversity in shaping portfolio performance and the need for differentiated pricing strategies. Costs of clients can vary greatly, depending on demand pattern and PV coverage. This insight has practical implications for contract design and stakeholder engagement for BRP's, as they enable contracts that reward predictable clients with low costs.

Ultimately, cooperative game theory provides a tool through which BRPs can foster trust, transparency, and long-term collaboration. As energy systems evolve toward decentralization, such approaches will be vital in aligning economic efficiency with social equity.

APPENDIX A

Appendix

A.1 Budget imbalance of risk problem caused by changing of indices in α -tail

Under the assumption that none of the worst case scenarios has a cost of 0, and that agents always have a higher imbalance than the aggregated imbalance, such a change incurs a subsidy of:

$$Tax_i(N) = \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} (J_{t,N \setminus \{i\}}) - \sum_{t \in \mathcal{T}_\alpha(J_{t,N \setminus \{i\}})} (J_{t,N \setminus \{i\}}) \quad (\text{A.1})$$

Proof. Suppose t_1 and t_2 are the time periods that are switched out. Except for t_1 and t_2 , the α -tail sets will be the same: $\mathcal{T}_\alpha(J_{t,N}) \setminus t_1 = \mathcal{T}_\alpha(J_{t,N \setminus \{i\}}) \setminus t_2$. t_1 is the time that enters the grand coalition set $t_1 \in \mathcal{T}_\alpha(J_{t,N})$ and t_2 only exists in the set without the client $t_2 \in \mathcal{T}_\alpha(J_{t,N \setminus \{i\}})$.

Marginal contribution can be broken up into marginal contribution per $t \in \mathcal{T}_\alpha(J_{t,N})$. Under the assumption that agents always have a higher imbalance than the aggregated imbalance, all agents j that does not change an index in the tail $t_1 \in \mathcal{T}_\alpha(J_{t,N \setminus \{j\}})$ will have a marginal contribution to the time t_1 of $M_{i,t_1} = J_{t_1,N} - J_{t_1,N \setminus \{i\}}$. If this is true for all clients, then $\sum_{i \in N} M_{i,t_1} = v_{t_1}$.

If $t_1 \notin \mathcal{T}_\alpha(J_{t,N \setminus \{i\}})$ and a client increases the cost at time t_1 causing it to enter $\mathcal{T}_\alpha(J_{t,N})$, their marginal contribution based on the time t_1 will be $M_{i,t_1} = J_{t_1,N} - J_{t_2,N \setminus \{i\}}$, as t_2 will be replaced by t_1 , instead of $J_{t_1,N} - J_{t_1,N \setminus \{i\}}$, the contribution to costs in t_1 for all other clients. This provides a discount that has to be paid as a subsidy: $Tax_i = J_{t_1,N \setminus \{i\}} - J_{t_2,N \setminus \{i\}}$ which is a reflection of the imbalance that was "absorbed" by changing the tail set. The same is true if the client decreases the cost of t_2 , causing t_1 to enter the tail. The marginal contribution will be based on how much the client decreased costs $J_{t_1,N} - J_{t_2,N \setminus \{i\}}$ instead of their costs in the relevant time $J_{t_1,N} - J_{t_1,N \setminus \{i\}}$.

This generalizes to: $Tax_i(N) = \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} (J_{t,N \setminus \{i\}}) - \sum_{t \in \mathcal{T}_\alpha(J_{t,N \setminus \{i\}})} (J_{t,N \setminus \{i\}})$

Example: A game with two periods and $\alpha = 0.5$: $T = \{1, 2\}$, has a coalition S with costs $J_{t,S} = \{3, 2\}$, causing the tail $\mathcal{T}_{0.5}(J_{t,S}) = \{1\}$ and $v(S) = 3$. Another agent has the costs $J_{t,\{i\}} = \{0.5, 2\}$, and joins S to create the coalition $N = S \cup \{i\}$, leading to costs $J_{t,N} = \{3.5, 4\}$, a new tail $\mathcal{T}_{0.5}(J_{t,N}) = \{2\}$, and $v(N) = 4$. The marginal contribution of agent i is $M_i = 1$, but their valuation is $v_i(N) = 2$, so they must be subsidized $M_i = v_i(N) - Tax_i(N) \Rightarrow Tax_i(N) = -1$, which is the cost difference of the new tail

and the old tail. The same happens if the tail is changed by decreasing cost instead of increasing it, e.g. if $J_{t,S} = \{3, 2\}$ and $J_{t,\{i\}} = \{-2, 0\}$, then $M_i = -1$ while $v_i(N) = 0$, needing a subsidy of $Tax_i = -1$.

As $\sum_{t \in \mathcal{T}_\emptyset(J_{t,N})} (J_{t,N \setminus \{i\}}) \leq \sum_{t \in \mathcal{T}_\emptyset(J_{t,N \setminus \{i\}})} (J_{t,N \setminus \{i\}})$, the taxes will always be negative, indicating a subsidy.

A.2 WMAPE dependent on scenarios

U	WMAPE
1	43.1%
5	41.2%
10	39.9%
15	40.2%
20	40.2%
25	37.9%
30	37.2%

Table A.1: In-sample WMAPE dependent on number of price scenarios, W=5

W	WMAPE
1	36.65%
2	38.00%
3	38.30%
4	37.90%
5	37.20%
6	36.80%
7	36.60%
8	36.40%
9	36.60%

Table A.2: In-sample WMAPE dependent on number of demand scenarios, U=30

A.3 22 Client Cost Timing

Allocation	Characteristic function time [s]	Allocation time [s]	Total time [s]
Shapley	71.0787	80.6093	151.6880
VCG	1.12372	0.0000096	1.12373
VCG (budget balanced)	1.12372	0.0014460	1.12516
Gately	1.12372	0.0000136	1.12373
Gately (interval)	1.12372	0.0093322	1.13305
Full cost	1.12372	0.0009508	1.12467
Reduced cost	1.12372	0.0007000	1.12442

Table A.3: Computation Times for Coalitions with 22 Clients, cost problem

A.4 Risk problem timing

Allocation	Characteristic function time [s]	Allocation time [s]	Total time [s]
Shapley	2.19479	0.008586	2.20338
VCG	0.083823	0.0000043	0.0838273
VCG (budget balanced)	0.083823	0.0012108	0.0850338
Gately	0.083823	0.0000067	0.0838297
Nucleolus	2.19479	65.6037	67.7985
Full cost	0.083823	0.0000108	0.0838338

Table A.4: Computation Times for Coalitions with 12 Clients, risk problem

Allocation	Characteristic function time [s]	Allocation time [s]	Total time [s]
Shapley	76.3944	82.7017	159.0960
VCG	0.20012	0.0000157	0.2001357
VCG (budget balanced)	0.20012	0.0032938	0.2034138
Gately	0.20012	0.0000126	0.2001326
Full cost	0.20012	0.0000158	0.2001358

Table A.5: Computation Times for Coalitions with 22 Clients, risk problem

A.5 Preferred allocations of clients

Client	Cheapest Allocation	Min €/MWh	Most Expensive Allocation	Max €/MWh
A	Gately 15Min	18.80	Reduced Cost	20.08
G	Reduced Cost	17.79	VCG BB	19.07
I	Reduced Cost	35.82	Gately 15Min	38.52
Q	VCG BB	22.79	Gately 15Min	24.78
S	VCG BB	20.38	Gately Full Period	21.95
H	Reduced Cost	40.61	Gately 15Min	43.99
L	Uniform Price	34.82	Gately 15Min	37.23
K	VCG BB	16.65	Uniform Price	18.61
U	VCG BB	30.08	Gately 15Min	32.02
D	VCG BB	28.98	Gately 15Min	32.61
P	VCG BB	6.79	Reduced Cost	26.82
M	VCG BB	64.19	Uniform Price	70.99
Y	Reduced Cost	71.79	Gately 15Min	79.05
O	VCG BB	57.60	Gately 15Min	64.50
T	VCG BB	38.38	Uniform Price	44.74
E	Reduced Cost	56.90	Gately 15Min	62.05
J	VCG BB	49.50	Gately 15Min	54.74
V	VCG BB	19.65	Reduced Cost	27.04
F	Reduced Cost	68.42	Gately 15Min	74.20
N	VCG BB	54.94	Uniform Price	59.82
W	VCG BB	38.96	Gately 15Min	42.30
X	VCG BB	42.19	Reduced Cost	50.87

Table A.6: Cheapest and Most Expensive Allocations per Client

A.6 Nomenclature

Nomenclature

α	Risk confidence level for CVaR.
β	Balancing function in bondareva-shapley theorem
δ'	Agents in the cost game that are not in δ^+ or δ^-
δ^+	Imbalance causers in the cost game
$\Delta_{w,t}^+$	Positive Imbalance of clients
δ^-	Imbalance reducers in the cost game
$\Delta_{w,t}^-$	Negative Imbalance of clients
$\Delta_{t,S}^{Real}$	Realized imbalance of coalition
$\Delta_{w,t}$	Imbalance of clients
$\epsilon(S, x)$	Excess of coalition S under allocation x : $\sum_{i \in S} x_i - v(S)$.
λ_t^{DA}	vector of spot prices
λ_t	vector of imbalance spread
$\lambda_{u,t}$	Imbalance spread in scenario
$\mathcal{T}_\alpha(J_{t,S})$	Indexes t of α worst scenarios
$\text{CVaR}_\alpha(J)$	Conditional Value-at-Risk at level α .
$\text{VaR}_\alpha(J)$	Value-at-Risk at level α .
μ	Mean of normal distribution
Φ	Shapley value
$\phi_i(v)$	Shapley-value allocation for client i in game v .
π_t	Day-ahead market price at time t .
Ψ	Set of coalitions that have assigned a locked excess when calculating nucleolus
ψ_S	Excess of coalition in nucleolus
σ	Standard deviation of normal distribution

- $\sigma_{u,t}$ Dominant direction in scenario (1 if positive 0 if negative)
- $\theta(x)$ Vector of excesses of all coalitions with respect to allocation x
- Θ_i Set of types for player i
- d^* Equalized and minimum possible propensity to disrupt (Gately equilibrium).
- $d_i(x)$ Propensity to disrupt of i under allocation x .
- d_t Dominant imbalance direction at t , $d_t \in \{\pm 1\}$.
- $i \in N$ Clients, members of the grand coalition.
- $J_{t,S}$ Cost of imbalance for coalition S at time t .
- $k^{\text{PVScaling}}$ Ratio between max production in data and PV plant capacity
- k_t^{uni} Variable to ensure budget balance for uniform price allocation under non-dominant direction imbalance
- $L(x_i, x_i^{\text{vick}})$ Distance function between two allocations
- M_i Marginal contribution of client i : $v(N) - v(N \setminus \{i\})$.
- N The grand coalition, Set of all clients.
- $P_{t,i}^{*DA}$ Optimal bid of client.
- $P_{t,i}^{\text{DA}}$ Day-ahead bid of client i at time t .
- $P_{t,i}^{D,\text{Real}}$ Realized consumption of client i at time t .
- $P_{w,t,i}^D$ Expected consumption of client i at time t in scenario w .
- $P_{t,i}^{G,\text{Real}}$ Realized generation (PV) of client i at time t .
- $P_{w,t,i}^G$ Expected generation of client i at time t in scenario w .
- $P_{\text{case}}^{\text{max}}$ Capacity of PV plant, 14 MW
- $P_{\text{data}}^{\text{max}}$ Max value of production in data
- $S \subseteq N$ A coalition, i.e. any subset of clients.
- $t \in T$ Index for time intervals.
- T Set of discretized time intervals (e.g. 15-minute periods).
- $\text{Tax}_i(N)$ VCG tax (externality) of agent i in grand coalition N .
- $u \in U$ Index for price-scenario realizations.

U	Set of imbalance-price scenarios.
v	A game, also denoted u or w sometimes
$v(S)$	Characteristic function (maximum gain/minimum cost) of coalition S in game v .
$v_{\text{cost}}(S)$	Characteristic function (total cost) of coalition S , $\sum_{t \in T} J_{t,S}$.
$v_{\text{risk}}(S)$	Game-value as CVaR of the cost series $\{J_{t,S}\}$, $\text{CVaR}_\alpha(\{J_{t,S}\})$.
$w \in W$	Index for demand-forecast scenarios.
W	Set of demand-forecast scenarios.
x_i^S	Allocated payoff or payment to client i as part of coalition S . Positive allocations are payments.
x_i	Allocated payoff or payment to client i . Positive allocations are payments.
x_i^{vick}	unmodified VCG allocation
z_i	Relative cost or CVaR. Allocated cost as part of grand coalition vs singleton
BIC	Bayesian Incentive Compatible
DSIC	Dominant Strategy Incentive Compatible
PVCover	Ratio of produced power to consumer power
WMAPE	Weighted Mean Average Percentual Error

A.7 Code

Code can be found in the repository: <https://github.com/AsmusWE/Sharing-the-Gains-of-Aggregation-Cooperative-Redistribution-of-Imbalance-Cost-Savings>

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